Policy News and Risk Premia in Monetary Policy Announcements*

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Abstract

Interest rate surprises around FOMC announcements mix pure monetary policy (MP) shocks with central bank information (CBI), Fed response to news (FRN), and risk premium (RP) shocks. Most high-frequency studies assume that policy surprises, measured in a short window, contain no RP shock. I relax this assumption and identify all four shocks jointly with sign and magnitude restrictions derived from an affine asset pricing model. Omitting FRN from the identification contaminates the MP shock, which then has smaller estimated effects on output and prices than the jointly identified one. Omitting RP costs little: the MP shock then contains almost no RP, and about a tenth of what is labeled CBI is RP. However, the RP shock is important on its own: it drives about 60% of the variation in ten-year yield surprises. It lowers the ten-year yield mainly through the term premium and raises housing starts and investment.

Keywords: monetary policy shocks, risk premium, central bank information effect, Fed response to news, stock-bond co-movement

JEL Classification: E52, E58

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1 Introduction

Almost every aggregate variable in macroeconomics is endogenous: policy responds to the state of the economy and the state of the economy responds to policy. This makes the estimation of the causal effects of monetary policy challenging, and requires a method that can identify policy moves orthogonal to the state of the economy: structural policy shocks. Among the methods proposed in the literature, high-frequency identification is one of the most widely used. This approach measures the change in the price of a financial asset in a tight window around the policy announcement (monetary policy surprise, MPS) and uses it as an instrument for the causal effects of monetary policy (Kuttner, 2001; Gürkaynak et al., 2005; Nakamura and Steinsson, 2018). The approach assumes that asset prices move within a tight (usually 30-minute) window around the announcement only because of its unanticipated content, so that the surprise measures the part of the policy change not driven by the state of the economy.

Despite being unanticipated, MPS is not exogenous: within the announcement window, the market surprise combines several distinct structural innovations. To see this, consider the policy surprise in the two-period bond yield, $\text{MPS}_t \equiv y_t^{(2)} - \mathbb{E}^m_{t-\Delta}\big[y_t^{(2)}\big]$, where $t-\Delta$ and t are the moments just before and just after the announcement, and $\mathbb{E}^m_s[\cdot]$ is the market's expectation conditional on the information it has at moment s. The two-period yield $y_t^{(2)} = \frac{1}{2}\big(i_t + \mathbb{E}^m_t[i_{t+1}]\big) + TP_t^{(2)}$ is the average of the current and the expected next-period policy rate plus a term premium, the compensation investors require for holding the two-period bond instead of rolling over one-period bonds. To keep the example minimal, the announcement carries no news about the next period's rate, $\mathbb{E}^m_t[i_{t+1}] = \mathbb{E}^m_{t-\Delta}[i_{t+1}]$.[1] Substituting the two-period yield into the definition of MPS_t and using the law of iterated expectations, $\mathbb{E}^m_{t-\Delta}\big[\mathbb{E}^m_t[i_{t+1}]\big] = \mathbb{E}^m_{t-\Delta}[i_{t+1}]$, shows that the policy surprise is half the surprise in the current rate plus the surprise in the term premium:

$$\text{MPS}_t = \tfrac{1}{2}\big(i_t - \mathbb{E}^m_{t-\Delta}[i_t]\big) + \underbrace{\tfrac{1}{2}\big(\mathbb{E}^m_t[i_{t+1}] - \mathbb{E}^m_{t-\Delta}[i_{t+1}]\big)}_{=0 \text{ by assumption}} + TP_t^{(2)} - \mathbb{E}^m_{t-\Delta}\big[TP_t^{(2)}\big].$$

Suppose the Fed sets the policy rate in response to a macro factor g_t, $i_t = a^{cb}\,\mathbb{E}^{cb}_t[g_t] + \varepsilon_t^{MP}$, where $\mathbb{E}^{cb}_t[\cdot]$ is the Fed's expectation conditional on its own information at the announcement, which can include information the market does not have, and ε_t^{MP} is a deviation from the systematic part of the rule. Before the announcement, the market expects $\mathbb{E}^m_{t-\Delta}[i_t] =$

[1] Without this assumption, the middle term of the display below is half the revision of the expected next-period rate. If the news behind this revision also moves the current rate, it is already part of the MP, FRN, or CBI component of (1), and only the weights of these components change. News that moves only future rates, such as pure forward guidance, is the exception: (1) has no component for it.

$a^m\,\mathbb{E}^m_{t-\Delta}[g_t]$, where a^m is the market's perceived response coefficient, which can differ from the true coefficient a^{cb}. Adding and subtracting $a^{cb}\,\mathbb{E}^m_{t-\Delta}[g_t]$ splits the surprise in the current rate into three parts:

$$i_t - \mathbb{E}^m_{t-\Delta}[i_t] = \varepsilon^{MP}_t + (a^{cb} - a^m)\,\mathbb{E}^m_{t-\Delta}[g_t] + a^{cb}\big(\mathbb{E}^{cb}_t[g_t] - \mathbb{E}^m_{t-\Delta}[g_t]\big).$$

The policy surprise then decomposes into four pieces:

$$\mathrm{MPS}_t = \frac{1}{2}\Big[\underbrace{\varepsilon^{MP}_t}_{\text{MP}} + \underbrace{(a^{cb} - a^m)\,\mathbb{E}^m_{t-\Delta}[g_t]}_{\text{FRN}} + \underbrace{a^{cb}\big(\mathbb{E}^{cb}_t[g_t] - \mathbb{E}^m_{t-\Delta}[g_t]\big)}_{\text{CBI}}\Big] + \underbrace{TP^{(2)}_t - \mathbb{E}^m_{t-\Delta}\big[TP^{(2)}_t\big]}_{\text{RP}}. \tag{1}$$

The first term is the *monetary policy* (MP) component, an exogenous deviation ε^{MP}_t from the systematic policy rule (Ramey, 2016). The second is a *Fed response to news* (FRN) component: the market's perceived response coefficient a^m differs from the true a^{cb}, so the term $(a^{cb} - a^m)\,\mathbb{E}^m_{t-\Delta}[g_t]$ surprises the market, although it is a function of information the market already had; the surprise is therefore predictable *ex post* from public pre-announcement information (Bauer and Swanson, 2023a,b; Swanson, 2024). Unlike the MP and CBI components, it is not an innovation dated t: it is predetermined, and it is a shock only in the sense that the market does not anticipate it. The third is a *central bank information* (CBI) component: the part of the policy surprise due to the Fed revealing a different outlook from the market's pre-announcement expectations (Romer and Romer, 2000; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020).

The fourth term is the change in the term premium within the window. Much of the literature that uses monetary policy surprises assumes that announcements do not move the risk premium in the futures contracts from which the surprises are built, either because the premium varies mainly at business-cycle frequencies or because it is thought to matter little at very short maturities (Miranda-Agrippino and Williams, 2026). Exceptions include studies that identify a risk premium or risk-appetite component of announcement surprises (Cieslak and Schrimpf, 2019; Bauer et al., 2026) and Miranda-Agrippino and Williams (2026), who relax the assumption for the shortest-maturity surprise. I relax it in a system that also identifies the CBI and FRN shocks and add a fourth shock, the *risk premium* (RP) shock: a change in the risk premium, the compensation investors require for bearing risk, that comes from a change in their willingness to bear risk or in the amount of risk they perceive (Bauer et al., 2023; Kroencke et al., 2021; Boehm and Kroner, 2024).

The risk premium is neither small nor constant in long yields and in equities. Cieslak and Schrimpf (2019) and Cieslak and Pang (2021) show that it moves within the announcement window, Bernanke and Kuttner (2005) attribute most of the equity response to the federal

funds rate surprise to news about expected excess returns (the equity premium), Hanson and Stein (2015) attribute the response of distant real forward rates to term premia, and Kroencke et al. (2021) and Boehm and Kroner (2024) identify components of the response of risky asset prices to FOMC announcements that are orthogonal to changes in short- and long-term yields. News about the policy rate also moves risk premia, as part of the transmission of the MP, CBI, and FRN shocks (Pflueger and Rinaldi, 2022; Borio and Zhu, 2012); the RP shock is the premium movement that no rate news causes. An announcement can move the premium without news about the policy rate by resolving uncertainty (Bauer et al., 2022; Ai and Bansal, 2018), or through news about asset purchases, which moves long yields without moving the expected short rate (Swanson, 2024).

Each component of (1) is driven by a single underlying shock. Because the FRN component is predetermined, it is orthogonal to the MP, CBI, and RP components, which are innovations at the announcement, so four mutually orthogonal shocks are present in the policy announcement window. High-frequency identification targets the MP shock. The literature on the information effect and on the Fed response to news treats the CBI and FRN components as sources of bias when the surprise is used to measure ε_t^{MP} (Jarociński and Karadi, 2020; Bauer and Swanson, 2023a). Each of these studies separates one of the two components and leaves the others unidentified. This paper asks where a non-policy shock that an identification omits ends up, and whether it contaminates the policy shock.

No existing identification accommodates all four. The Fed response to news and the risk premium have been studied in two strands of literature that developed separately. Jarociński and Karadi (2026) (hereafter JK) identify MP, CBI, and FRN shocks using sign and magnitude restrictions on three high-frequency observables: the Bauer and Swanson (2023b) fitted value and residual, the components of the policy surprise that are respectively predictable and unpredictable from pre-announcement information, and the S&P500 surprise. JK assume away the risk-premium channel. Cieslak and Schrimpf (2019) (hereafter CS) identify MP, CBI, and RP shocks from the joint window behavior of equities and short- and long-maturity Treasury yields, and assume away the FRN channel instead. The CS risk premium shock captures windows in which stocks and yields fall together, with long yields falling by more than short ones: investors shift from risky assets toward safer Treasuries, pushing bond prices up and yields down, especially at the long end. Each scheme rules out precisely the shock the other treats as important.

When a structural shock is omitted, its variation does not disappear. It is absorbed by the shocks that remain, which then become contaminated. In the announcement window, the FRN shock moves asset prices in the same directions as the MP shock, and the RP shock moves stocks and yields together, as the CBI shock does. A system without the FRN shock

can therefore contaminate the policy shock directly, and a system without the RP shock can contaminate the information shock and, through it, the policy shock. Which component is removed can change the estimated effects of monetary policy: Bauer and Swanson (2023b) remove the response to news from the policy surprise and find effects on output and prices of the conventional sign, while Bauer et al. (2026) separate a risk-appetite shock from the rate surprise and find small and imprecisely estimated effects of the remaining rate shock. The contamination from missing CBI or FRN, when not identified together with the MP shock, has been documented (Jarociński and Karadi, 2020, 2026; Bauer and Swanson, 2023a). Swanson (2025) reviews this evidence. Regressions of private forecast revisions on policy surprises, the main evidence for an information effect, omit the public news released before the announcement, which moves both the surprise and the revisions (Bauer and Swanson, 2023a). Once this news is controlled for, the evidence for an information effect disappears. He conjectures that the occasional information shock that Jarociński and Karadi (2020, 2026) identify is probably the risk-appetite, or non-yield, shock of Boehm and Kroner (2024). There is no evidence on the contamination that involves the risk premium shock in a system that also identifies the Fed response to news and information shocks. Bauer and Swanson (2023a) show that the FRN component accounts for a substantial share of the variance of the policy surprise, though the size of the channel remains debated (Miranda-Agrippino and Williams, 2026); the CS system contains no observable that isolates that component, so its variation must be absorbed by some combination of the MP, CBI, and RP shocks. Likewise, the risk premium shock is a large component of announcement surprises (Cieslak and Schrimpf, 2019) and is likely absorbed by some combination of the shocks considered in JK.

Neither system can say where the missing shock's variation ends up, or how large it is. Answering that requires a system in which all four shocks are identified together. This paper provides the first joint identification of MP, CBI, FRN, and RP at FOMC announcements. I propose a unified sign- and magnitude-restriction framework, henceforth JKCS, that combines the Jarociński–Karadi and Cieslak–Schrimpf systems, so that each strand's shock is identified in the presence of the other's. JKCS retains the Bauer–Swanson decomposition of the policy surprise into a residual (RES) and a fitted value (FIT), which is what separates the MP shock from the FRN shock; keeps the high-frequency S&P500 surprise, which separates the central bank information shock (Jarociński and Karadi, 2020); and adds the ten-year Treasury yield as a fourth observable, which supplies the maturity dimension required to separate the risk premium shock from the other shocks that move stocks and long yields together. The resulting 4×4 structural system is set-identified by sign and magnitude restrictions of the kind used in JK and CS individually, derived from a simple affine asset pricing model that brings the two frameworks into one setup, together with an external

correlation restriction that disciplines the risk premium shock with the daily risk-appetite index of Bauer et al. (2023). Signs alone separate neither the MP shock from the FRN shock nor the CBI shock from the RP shock: magnitude restrictions on the RES and FIT responses separate the first pair, and magnitude restrictions across maturities separate the second. The JK and CS observables are combinations of the JKCS observables, so on the same announcement windows every JK and CS shock is an exact combination of the four JKCS shocks, and the share of its variance across announcements that comes from each JKCS shock is measured directly.

I estimate the three systems (JKCS, JK, CS) on the same sample of 382 announcement windows from January 1991 to March 2025: FOMC statements, including those after unscheduled meetings, and press conferences. The March 2020 announcements are excluded from the identification. Outside the window, I trace the effects of the shocks with daily local projections and a monthly Bayesian VAR.

Two results answer the question, and a third describes the risk premium shock itself. First, omitting the Fed response to news shock contaminates the policy shock. Without the Fed response to news shock, about a quarter of the variance of the policy shock is response to news, the direction that Section 2.6 and Bauer and Swanson (2023a) predict, and the estimated effects of the policy shock on output and prices are smaller than those of the jointly identified policy shock. Without the risk premium shock, the policy shock contains almost no risk premium variation. Bauer et al. (2026) find instead that a rate shock separated from a risk-appetite shock has small and imprecisely estimated effects; one explanation is that their system identifies neither a CBI nor an FRN shock, so both are absorbed by their two shocks.

Second, the cost of omitting the risk premium shock is small and falls on the information shock: about a tenth of the variance of the JK information shock is risk premium variation, and a similar share is monetary policy. The risk premium shock identified here moves the long end of the yield curve, while the shock in the conjecture of Swanson (2025) is the non-yield shock of Boehm and Kroner (2024), a move in stock prices and the dollar orthogonal to the yield curve. The risk premium content of the JK information shock is therefore term-premium variation and does not test that conjecture.

Third, the risk premium shock is important on its own: it is a long-end phenomenon that accounts for about 60% of the variance of the ten-year yield surprise within announcement windows and for almost none of the two-year yield or equity surprises. The identifying restrictions do not impose these shares: they require the risk premium shock to move the ten-year yield by more than the two-year yield and by more than the policy shock does, but

they leave unrestricted how much of each surprise it explains.[2] Omitting the Fed response to news shock does not affect the risk premium shock. In daily local projections the risk premium shock lowers the ten-year yield mainly through the term premium of Adrian et al. (2013), with no reaction from the expectations component after the announcement day. At the monthly frequency, housing starts and investment rise after it.

For applied work the implication is asymmetric: a policy shock identified without the risk premium shock contains almost none of it, while a policy shock identified without the Fed response to news shock mixes monetary policy with the Fed's predictable response to news. Surprises in long yields are a different matter: about 60% of their window variance is the risk premium shock, which in this system also contains news about asset purchases.

Section 2 develops an affine asset pricing model of the announcement window from which the sign and magnitude restrictions are derived. Section 3 introduces the three systems and their restrictions, and states the three comparability choices that make the shocks of the three systems measurable on the same windows, so that each JK and CS shock can be written as a combination of the four JKCS shocks. Section 4 describes the data. Section 5 reports what each shock explains within the announcement window, on average and announcement by announcement, and how the decomposition changes relative to JK and CS. Section 6.1 asks whether the four shocks move daily financial variables in the directions their labels predict. Section 6.2 follows with the reaction of macro variables at the monthly frequency, and Section 7 concludes.

2 An illustrative model of the four shocks

The identification in Section 3 rests on sign and magnitude restrictions on four announcement-window observables: the predictable and unpredictable parts of the policy surprise, the S&P500 surprise and the ten-year yield surprise. This section derives those restrictions in a simple asset-pricing model. I reformulate the Cieslak and Schrimpf (2019) affine model and add a mechanism their model does not have: the market underestimates how strongly the Fed responds to news about the economy, which makes part of the policy surprise, the FRN component, predictable. I then derive the window responses of bond yields and stock prices to this component and to the MP, CBI and RP shocks.

[2] Across the identified set, the 5th and 95th percentiles of the risk premium share of the ten-year yield surprise are 43 and 74%. The shock remains the largest contributor to that surprise in each of the seven configurations of Section 5, including those without the external correlation restriction and without the restriction relative to the policy shock. With every restriction on its column dropped except the ten-year sign that normalizes it, the bands widen and the other three shocks correlate at 0.97 or more with their baseline counterparts.

2.1 State vector and dynamics

Suppose that the economy is described by the state $F_t = (g_t,\ i_t,\ x_t)'$, where g_t is the macro factor, i_t the short rate, and x_t the price-of-risk factor. The macro factor is a summary measure of economic activity to which the Fed responds, and it also sets the expected growth of dividends (Appendix 8.2).[3] The price-of-risk factor can be read as investors' risk aversion: a rise in x_t is a fall in risk appetite, as in the risk-off episodes in which stock prices and Treasury yields fall together (Cieslak and Schrimpf, 2019). With all state variables in deviations from their unconditional means, the macro and risk factors follow AR(1) processes, and the policy rate is a persistent process that responds to the same-period macro factor with a fixed coefficient a^{cb}, the *true* response coefficient:

$$g_{t+1} = \phi_g g_t + \sigma_g \varepsilon^g_{t+1}, \qquad x_{t+1} = \phi_x x_t + \sigma_x \varepsilon^x_{t+1},$$
$$i_{t+1} = \phi_i i_t + a^{cb} g_{t+1} + \sigma_i \varepsilon^i_{t+1},$$

with $\phi_g, \phi_i, \phi_x \in (0,1)$ and $\varepsilon_{t+1} = (\varepsilon^g_{t+1}, \varepsilon^i_{t+1}, \varepsilon^x_{t+1})' \sim N(0, I_3)$ i.i.d.; ε^i is the MP shock.

The market forms expectations of the policy rate under a misspecified model in which the response coefficient is a known constant $a^m < a^{cb}$, the *perceived* response coefficient (Bauer and Swanson, 2023a), so the gap in coefficients is $\boldsymbol{b} \equiv a^{cb} - a^m > 0$. Under $\mathbb{E}$ the coefficient is the true a^{cb}, which generates the data, and under $\mathbb{E}^m$ it is a^m, which generates prices and forecasts. The persistent *ex post* predictability of policy surprises documented by Bauer and Swanson (2023a) supports this assumption.[4]

The model makes two simplifications. First, the macro factor does not respond to policy: neither i_t nor ε^i enters its equation (Cieslak and Schrimpf, 2019). A policy shock therefore changes the interest rates at which investors discount future dividends but not the dividends they expect, which separates news about cash flows from news about discounting in the announcement window. Allowing policy to move the macro factor moderately would not change the qualitative results (Cieslak and Pang, 2021). Second, the policy rate responds to the macro factor and not to expected inflation.

2.2 SDF and pricing

I start from the general pricing principle that for an asset with a price of one dollar today, the expected discounted payoff must equal the price, $\mathbb{E}^m_t[M_{t+1}R_{t+1}] = 1$, where R_{t+1} is the gross

[3] Cieslak and Schrimpf (2019) additionally include expected inflation in F. The qualitative implications I derive from my model are similar to those obtained in their paper.

[4] A rational agent would learn a^{cb} from the data, and Bauer et al. (2024) document that perceived policy rules do adjust in response to policy surprises. What the model needs is only that a^m not change at the announcement itself; variation between announcements would not change any of the model's implications.

return, the payoff in dollars next period, and M_{t+1} is the stochastic discount factor (SDF). In a consumption-based model the SDF is the intertemporal marginal rate of substitution, high in states where consumption is scarce: an asset paying badly in exactly those states must compensate its holder with an extra expected return, the risk premium.

Following Cieslak and Schrimpf (2019) and the affine no-arbitrage literature (Duffie and Kan, 1996; Lettau and Wachter, 2011; Koijen et al., 2017), I specify the SDF directly rather than derive it from a utility function. Assume the log SDF, $m_{t+1} = \ln(M_{t+1})$, is linear in next period's shocks,

$$m_{t+1} = -i_t - \tfrac{1}{2}\lambda_t'\lambda_t - \lambda_t'\varepsilon_{t+1},$$

where λ_t is known at date t and the intercept follows from the definition of the short rate (Appendix 8.1). This form of SDF holds whenever log marginal utility is linear in conditionally normal shocks. With power utility and log-normal consumption growth it holds with a constant λ. Time-varying risk aversion (habit models) or uncertainty (long-run risk models) makes λ_t time-varying (Cieslak and Pang, 2021).

The vector λ_t is called the price of risk: each entry is the extra expected return that investors require per unit of exposure to the corresponding shock, so a shock is compensated, that is *priced*, if and only if its entry is nonzero. I set, following Cieslak and Schrimpf (2019), $\lambda_t = (\lambda_{gx}x_t,\ 0,\ 0)'$ with $\lambda_{gx} > 0$: investors are compensated for growth risk alone, and the price they charge for it moves with x_t.

The first restriction follows Cieslak and Schrimpf (2019), who price only shocks to expected growth and expected inflation, and the consumption-based reading of the SDF, under which it moves with news about consumption (Cochrane, 2009). There is no consumption in the model, but the macro factor carries such news, and neither ε^i nor ε^x moves it.

The second restriction makes risk premia time-varying: the higher x_t, the more bad news about the macro factor raises the SDF, and the higher the expected return investors require on assets that pay badly after such news, such as stocks. Were the price of risk constant, premia would be constant too and the RP shock, which moves the term premium of bonds and the equity premium of stocks (Section 2.4), would not exist. Empirical proxies for risk appetite (Bekaert et al., 2022; Bauer et al., 2023) are typically procyclical; here x is defined as the component of risk aversion independent of g, so its cyclical part is absorbed into the macro factor.

2.3 Information, timing, notation

Periods $t, t+1, \ldots$ index FOMC announcements, one per period. For simplicity, no shocks occur between two consecutive announcements: all period-t innovations are realized and

revealed at the period-t announcement. At each announcement the market observes the state and knows the model and every parameter except the true response coefficient a^{cb}. Information immediately before announcement t, at moment $t-\Delta$, contains all states dated $t-1$ and earlier. For any variable W, $\mathbb{E}^m_{t-\Delta}[W_t]$ and $\mathbb{E}^m_t[W_{t+1}]$ denote the market's forecasts just before and just after announcement t, and $\widetilde{W}_t \equiv W_t - \mathbb{E}^m_{t-\Delta}[W_t]$ is the announcement surprise.

2.4 Asset-price responses and identification

2.4.1 Bond yields

Under these assumptions the zero-coupon n-period bond yield is affine in the state, $y^{(n)}_t = A_n + B_n' F_t$ with loadings $B_n = (B^g_n, B^i_n, B^x_n)'$, as in the models of Cieslak and Schrimpf (2019) and Cieslak and Pang (2021). Each yield is the average of expected future short rates (the expectations-hypothesis component) plus a term premium. Only x moves the term premium, and setting the price of risk to zero ($\lambda_{gx} = 0$) makes it constant, so that the expectations hypothesis holds (Appendix 8.1.1).

The announcement moves the state and not the loadings, so the window response of each yield is $\tilde{y}^{(n)}_t = B_n' \widetilde{F}_t$, with $\widetilde{F}_t = (\sigma_g \varepsilon^g_t,\ \tilde{\imath}_t,\ \sigma_x \varepsilon^x_t)'$: the surprises in the macro and risk factors equal their shocks. The rate surprise is the policy rule minus the market's forecast of it, $\mathbb{E}^m_{t-\Delta}[i_t] = \phi_i i_{t-1} + a^m \phi_g g_{t-1}$, so $\tilde{\imath}_t = \boldsymbol{b}\,\phi_g g_{t-1} + a^{cb}\sigma_g \varepsilon^g_t + \sigma_i \varepsilon^i_t$: the Fed responds to the predictable part of the macro factor, $\phi_g g_{t-1}$, with a^{cb}, while the market expects a^m. The perceived coefficient also enters the loadings B^g_n and B^x_n, which are computed under the market's model. Substituting,

$$\tilde{y}^{(n)}_t = \underbrace{B^i_n\, \boldsymbol{b}\, \phi_g g_{t-1}}_{\text{FRN}} + \underbrace{\sigma_g \big(B^g_n + a^{cb}\, B^i_n\big)\varepsilon^g_t}_{\text{CBI}} + \underbrace{\sigma_i B^i_n\, \varepsilon^i_t}_{\text{MP}} + \underbrace{\sigma_x B^x_n\, \varepsilon^x_t}_{\text{RP}}.$$

MP, CBI and FRN move yields only through expected short rates, RP only through the term premium. For $n = 2$, the two-period yield of the example in the introduction, the display has the four components of equation (1), with $\mathbb{E}^{cb}_t[g_t] = g_t$ (the Fed observes g_t) and $\mathbb{E}^m_{t-\Delta}[g_t] = \phi_g g_{t-1}$; only the weights differ, because in the model the announcement also moves the expected next-period rate, which the example rules out. RP does not move the policy rate, the one-period yield ($B^x_1 = 0$). This single zero is the reason the RP shock can be invisible to a system built on the policy rate surprise and dominant in the ten-year yield, which is what Section 5 finds in the data.

The four shocks are mutually orthogonal: $\varepsilon^g, \varepsilon^i, \varepsilon^x$ are orthogonal by construction, and the FRN component $\boldsymbol{b}\,\phi_g g_{t-1}$ is known before the announcement, hence orthogonal to the

period-t innovations. It is a shock only in the sense that the market, which forecasts with a^m, does not anticipate it. Following Bauer and Swanson (2023a), the monetary policy surprise MPS, here the two-year yield surprise,[5] is decomposed into a component predictable from a vector Z_t of public information known before the announcement and a residual,

$$\text{MPS}_t = \underbrace{\hat{\beta}' Z_t}_{\text{FIT}_t} + \underbrace{\widehat{\varepsilon}_t}_{\text{RES}_t}. \tag{2}$$

In the model, where Z_t contains g_{t-1}, FIT is exactly the FRN term of the two-year yield surprise and RES contains the other three. The two enter the identification of Section 3 as separate observables.

The loadings are derived in closed form in Appendix 8.1. Under the normalization that $\varepsilon_t^i > 0$ is a tightening, $\varepsilon_t^g > 0$ good growth news, $\varepsilon_t^x > 0$ a rise in the price of risk, and FRN positive when g_{t-1} is above its mean, the sign of each yield response is the sign of its loading: $B_n^i > 0$ for MP and FRN, $B_n^g + a^{cb} B_n^i > 0$ for CBI, and $B_n^x \leq 0$ for RP. MP, CBI and FRN therefore raise yields at every maturity, and RP lowers them. B_n^i falls with maturity, while B_n^g and $|B_n^x|$ are hump-shaped.

All four shocks flatten the yield curve, but by affecting different parts of the curve: MP and FRN raise yields by more at the short end, CBI raises yields by more in the middle, while RP lowers them by more at the long end. The first three shocks act only on expected future short rates, and since $\phi_i < 1$, shorter rates move more than long ones; the RP shock moves the term premium, which accumulates with maturity. These orderings agree with the evidence summarized by Cieslak and Pang (2021): monetary shocks move the two-year yield about two to three times as much as the ten-year yield, the effect of real-activity news declines beyond maturities of two to three years, and a risk premium shock moves the ten-year yield more than twice as much as the two-year yield (Cieslak and Povala, 2015). Figure 10 in Appendix 8.1 traces the range of ϕ_i over which the CBI and RP orderings hold jointly.

2.4.2 Stocks

As in Lettau and Wachter (2011), Cieslak and Schrimpf (2019) and Cieslak and Pang (2021), the same SDF prices stocks; the joint window response of stocks and yields is what Jarociński and Karadi (2020) and Cieslak and Schrimpf (2019) use to separate monetary policy from other news. The log price–dividend ratio is affine in the same state, $pd_t = A_s + B_s' F_t$ with

[5] In the original paper, MPS is the first principal component of the first four eurodollar futures. I use the two-year yield to compare my identification directly with Cieslak and Schrimpf (2019), who also use it as an observable.

$B_s = (B_s^g, B_s^i, B_s^x)'$, and within the window dividends are predetermined, so the stock price response $\widetilde{\log P}_t = B_s' \widetilde{F}_t$ has the same four terms as the yield response, with B_s in place of B_n (Appendix 8.2). MP, CBI and FRN affect stocks through expected cash flows and short rates, RP only through the equity premium. Only $B_s^i < 0$ holds for all parameter values, so MP and FRN lower stock prices within the announcement window. CBI raises stock prices ($B_s^g + a^{cb}\, B_s^i > 0$) and RP lowers them ($B_s^x < 0$) provided a weighted average of the true and the perceived long-run response of the policy rate to real activity is below one (Appendix 8.2 gives the conditions). Estimates of the long-run response of the policy rate to the output gap range from 0.65 to 1.01 (Carvalho et al., 2021); at the calibration of Section 2.5, which takes the upper end, the weighted average is 0.91. Equity is exposed to growth risk while bonds hedge it (Cieslak and Pang, 2021), so RP lowers stock prices while yields fall.

2.5 Summary of model results

Table 1 collects the results of the model. Columns are the shocks. Panel A reports the sign of each window response: of the two-year yield surprise $\tilde{y}_t^{(N_2)}$ and its two parts, RES and FIT; of the S&P500 surprise $\widetilde{\log P}_t$; and of the ten-year yield surprise $\tilde{y}_t^{(N_{10})}$, where N_2 and N_{10} are the numbers of announcements in two and ten years. The two-year yield surprise is RES + FIT, so it is not a fifth observable. Panel B reports which of two responses to the same shock is larger in absolute value: that of RES or of FIT, and that of the two-year or of the ten-year yield, which determines the response of the yield-curve slope.

Table 1: Model-implied window responses

	MP (ε^i)	FRN ($\boldsymbol{b}\phi_g g_{t-1}$)	CBI (ε^g)	RP (ε^x)	
A. Sign of the response					
Two-year yield surprise, $\tilde{y}_t^{(N_2)}$	$+$	$+$	$+$	$(-)$	
$=$ RES, unpredictable part	$+$	*free*	$+$	$(-)$	
$+$ FIT, predictable part	*free*	$+$	*free*	*free*	
S&P500 surprise, $\widetilde{\log P}_t$	$-$	$-$	$+^*$	$-^{**}$	
Ten-year yield surprise, $\tilde{y}_t^{(N_{10})}$	$+$	$+$	$+$	$-$	
B. Larger response to the same shock					*separates*
RES or FIT	RES	FIT	RES	RES	MP from FRN
$\lvert\tilde{y}_t^{(N_2)}\rvert$ or $\lvert\tilde{y}_t^{(N_{10})}\rvert$	2Y	2Y	2Y†	10Y†	CBI from RP

Notes. Normalization: $\varepsilon^i > 0$, $\boldsymbol{b}\phi_g g_{t-1} > 0$, $\varepsilon^g > 0$, $\varepsilon^x > 0$. MP and FRN differ only in the RES and FIT rows; CBI and RP have opposite signs in every row of Panel A except FIT, and since the sign of a shock is a normalization, signs cannot separate them. The last column names the pair that each comparison separates. In the model FIT is exactly the FRN term; in the data FIT is an estimate. Most entries hold for any parameter values; some require additional conditions. *CBI requires $B_s^g + a^{cb}B_s^i > 0$ and **RP requires only $B_s^x < 0$; the first implies the second, and both hold for typical Taylor rule estimates. †These two entries imply conditions on persistence, and for a typical calibration they hold.

Signs separate four of the six pairs of shocks. They cannot separate MP from FRN, which both move stocks and yields in opposite directions, or CBI from RP, which both move them in the same direction. Panel B separates these two pairs. First, MP and FRN are proportional in every asset price and differ only in predictability, so no sign restriction on asset prices separates them; their curves in Figure 1 coincide. They are separated instead by which part of the surprise they move more, RES or FIT, which makes the Bauer–Swanson split a necessary feature of the identification. Second, RP is the only shock with a yields-*down*, stocks-*down* signature, but this is the signature of CBI reversed. Since the sign of a shock is a normalization, no sign restriction separates the two, and the maturity dimension is the only observable that isolates RP from CBI: CBI moves the two-year yield more than the ten-year yield and RP the reverse, so their curves in Figure 1 are not mirror images.

Figure 1 plots the model's window yield response to each shock against maturity, and the sign of the stock price response, at the following calibration: eight announcements per year, $\phi_g = 0.965$, $\phi_x = 0.9$, $\phi_i = 0.9$, $\gamma^{cb} = 1.01$ and $\gamma^m = 0.9\,\gamma^{cb}$, where $\gamma^{cb} \equiv a^{cb}/(1-\phi_i)$ and $\gamma^m \equiv a^m/(1-\phi_i)$ are the true and perceived long-run responses of the policy rate to the macro factor. The persistence of the macro factor follows the real factor of Cieslak and Povala (2015), and that of the price-of-risk factor the bond risk premium estimates reported

in Cieslak and Pang (2021). The policy-rate persistence and γ^{cb} follow the interest-rate smoothing and the long-run response to the output gap that Carvalho et al. (2021) estimate for 1987–2007. With $\gamma^{m} = 0.9\,\gamma^{cb}$, the stock price conditions of Appendix 8.2 hold for any γ^{cb} below about 1.1. The other parameters, $\lambda_{gx}\sigma_g$ and the shock volatilities, only scale the responses, so the curves, scaled by the two-year response, depend only on the three persistences and, for CBI, on γ^{m}/γ^{cb}.

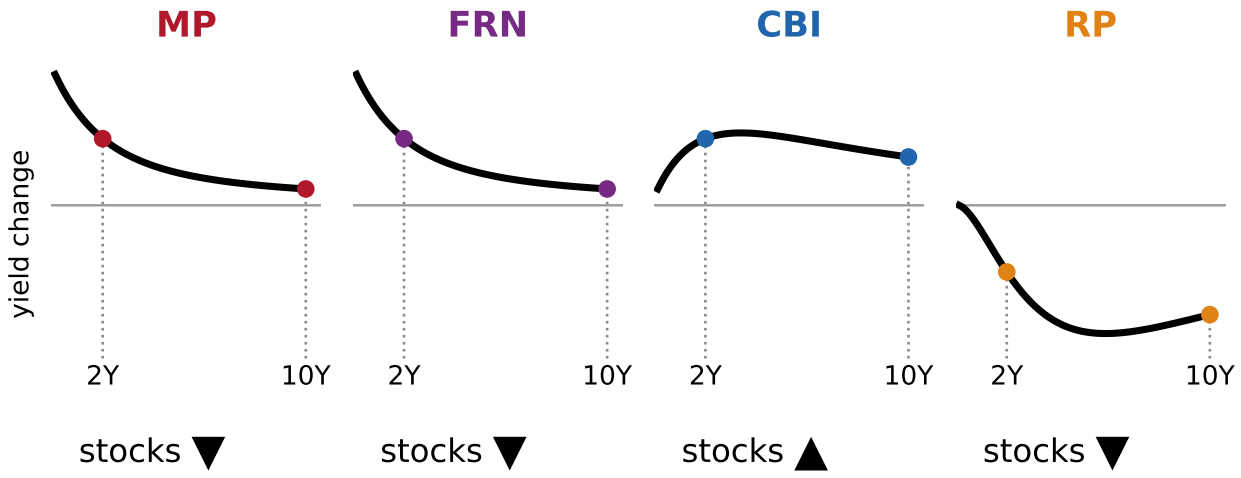


Figure 1: Model-implied window responses across the maturity curve

Notes. Each panel plots the model's window yield response to one shock against maturity, at the calibration of Subsection 2.5, scaled by the magnitude of the two-year response. Dotted lines and markers show the two- and ten-year responses, from which the slope is formed; the arrow below each panel gives the sign of the stock price response. Each entry in the last row of Table 1 holds when the ten-year marker lies below the two-year marker.

2.6 Remarks

First, the model is small by construction and has obvious limitations. What I take from it is only *qualitative*: the signs and comparisons in Table 1, which hold for any parameter values or under relatively mild conditions. The sizes of the four shocks are free parameters, so the model cannot say how the variance of an announcement divides among them, or, unless one of the two responses is zero, which of two shocks moves a given observable more.

Second, the variation of an omitted shock goes to whichever remaining shock moves asset prices the same way. Within the announcement window the FRN component carries no news about the economy (no ε_t^g), so it moves asset prices exactly as the policy shock does, and a system without FRN sends its variation into the policy shock and leaves the other shocks unchanged. Jarociński and Karadi (2026) find that without the FRN shock, the monetary policy shock implies a larger and more persistent response of the one-year

yield. In monthly data the two differ: the FRN component is high when growth has been high, and growth is persistent, so a rate increase driven by FRN tends to be followed by an expansion. A researcher who observes this pattern would attribute the rate increase to favorable information held by the Fed and classify it as a central bank information shock (Bauer and Swanson, 2023a).

Third, a system without a risk premium shock attributes its variation mainly to the information shock, with the sign reversed, because the two shocks have opposite signs in every observable (Section 2.4). Unlike the FRN component, the risk premium shock is absorbed only partly, because the two shocks also differ in the maturity ordering, and the degree of absorption depends on the variance of the risk premium shock relative to the other shocks, which the model does not restrict.

3 Identification

Let u_t denote the vector of observed reduced-form high-frequency surprises and ε_t the vector of structural shocks, linked by

$$u_t = B\,\varepsilon_t, \qquad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I, \tag{3}$$

so that $BB' = \Sigma_u$. Since B is not uniquely pinned down by Σ_u, identification requires additional restrictions. The three systems below differ in their observables and restrictions; Section 4 describes the surprises themselves and every other series the paper uses.

3.1 Jarociński–Karadi (2026)

JK identify three shocks, monetary policy (MP), Fed response to news (FRN), and central bank information (CBI), from $u_t = (\text{RES}_t,\ \text{FIT}_t,\ \Delta\text{S\&P500}_t)'$, with

$$\begin{pmatrix} \text{RES} \\ \text{FIT} \\ \Delta\text{S\&P500} \end{pmatrix} = \underbrace{\begin{pmatrix} + & a_{12} & a_{13} \\ a_{21} & + & a_{23} \\ - & - & + \end{pmatrix}}_{B^{\text{JK}}} \begin{pmatrix} \varepsilon^{MP} \\ \varepsilon^{FRN} \\ \varepsilon^{CBI} \end{pmatrix}, \tag{4}$$

subject to

$$a_{13} + a_{23} > 0, \qquad |a_{21}| < a_{11}, \qquad |a_{12}| < a_{22}. \tag{5}$$

Entries written a_{mj} carry no sign restriction ("free" in Table 2), and the signs on the diagonal are normalizations that label the shocks. The first condition in (5) makes a CBI shock move

MPS = RES + FIT and ΔS&P500 in the same direction. The other two require the MP shock to move RES more than FIT, and the FRN shock to move FIT more than RES.

JK additionally exploit heteroskedasticity between FOMC and non-FOMC events. When the variances of the shocks change across the two sets of events by different proportions, B is identified up to the order and signs of its columns (Rigobon, 2003), and the sign restrictions label the shocks. The sample here excludes speeches, testimonies and minutes releases (Subsection 4.1), so I use restrictions (4)–(5) only.

3.2 Cieslak–Schrimpf (2019)

CS identify three shocks, monetary policy (MP), growth, and risk premium (RP), from $u_t = (\Delta R_{2Y,t},\ \Delta\text{S\&P500}_t,\ \Delta R_{10Y,t})'$. Their growth shock moves stocks and yields together, as the CBI shock of JK does, and I refer to it as CBI throughout:

$$\begin{pmatrix} \Delta R_{2Y} \\ \Delta\text{S\&P500} \\ \Delta R_{10Y} \end{pmatrix} = \underbrace{\begin{pmatrix} + & + & - \\ - & + & - \\ + & + & - \end{pmatrix}}_{B^{\text{CS}}} \begin{pmatrix} \varepsilon^{MP} \\ \varepsilon^{CBI} \\ \varepsilon^{RP} \end{pmatrix}. \tag{6}$$

The RP shock is a change in the compensation investors require for bearing risk, such as a shift in sentiment or effective risk aversion, that moves equity prices and bond yields in the same direction, independently of news about the economy or the policy rate (Cieslak and Schrimpf, 2019). A positive RP shock lowers both, as government bonds become more valuable in flight-to-safety episodes. Such shocks are not specific to FOMC announcements, but they can also occur within the narrow FOMC window, because an announcement can affect policy uncertainty and carries information about risks to the outlook.

The RP column of (6) is the CBI column with every sign reversed, and the sign of a shock is a normalization, so the sign restrictions alone cannot separate the two shocks; the same is true of CBI and RP in JKCS. To distinguish them, CS exploit the maturity dimension of the yield curve through monotonicity restrictions:

$$|b_{2Y,RP}| < |b_{10Y,RP}|, \tag{7}$$

$$|b_{2Y,CBI}| > |b_{10Y,CBI}|, \tag{8}$$

$$|b_{2Y,MP}| > |b_{2Y,RP}|, \tag{9}$$

$$|b_{10Y,MP}| < |b_{10Y,RP}|, \tag{10}$$

where $b_{n,s}$ is the response of the n-maturity yield to shock s. CS note that (7) is essential

for separating the two shocks and that imposing the other restrictions changes their results only marginally; (9) and (10) compare MP with RP.

3.3 JKCS: joint identification

JKCS combines the two systems. It takes the RES/FIT/ΔS&P500 block from JK and adds the ten-year Treasury yield as a fourth observable:

$$\begin{pmatrix} \text{RES} \\ \text{FIT} \\ \Delta\text{S\&P500} \\ \Delta R_{10Y} \end{pmatrix} = \underbrace{\begin{pmatrix} + & c_{12} & + & - \\ c_{21} & + & c_{23} & c_{24} \\ - & - & + & - \\ + & + & + & - \end{pmatrix}}_{B^{\text{JKCS}}} \begin{pmatrix} \varepsilon^{MP} \\ \varepsilon^{FRN} \\ \varepsilon^{CBI} \\ \varepsilon^{RP} \end{pmatrix}, \tag{11}$$

where c_{mj} is the response of observable m to shock j; as in (4), entries written c_{mj} carry no sign restriction and the signs on the diagonal are normalizations. Table 1 reports the responses of five variables, but the two-year yield is not a fifth observable: under the measurement convention of Section 4 it equals RES + FIT exactly, so adding it would make Σ_u singular. A condition on the two-year yield is a condition on the sum of the first two rows of B.

The signs in (11) are those of Panel A of Table 1. As Subsection 2.5 shows, they separate four of the six pairs of shocks, but not MP from FRN, which move stock prices and yields alike, nor CBI from RP, whose signs are the same up to the normalization, as in CS.

Two sets of magnitude restrictions separate these two pairs. The first separates MP from FRN through the split of the policy surprise:

$$|c_{21}| < c_{11}, \quad |c_{12}| < c_{22}, \quad |c_{23}| < c_{13}, \quad |c_{24}| < |c_{14}|, \qquad |c_{12}| < c_{11}, \quad |c_{21}| < c_{22}. \tag{12}$$

That is, MP, CBI and RP move RES more than FIT and FRN moves FIT more than RES; RES responds more to MP than to FRN, and FIT more to FRN than to MP. All six hold in the model, where FIT is exactly the FRN term, and, with the signs, they imply that each shock moves the two-year yield in the same direction as the ten-year yield.

The second set separates CBI from RP through the maturity dimension, as in CS: CBI moves the two-year yield more than the ten-year yield and RP the reverse. I impose the same ordering on MP and FRN as well, as the model predicts, together with the two comparisons of MP with RP in (9)–(10):

$$|c_{1j} + c_{2j}| > |c_{4j}| \;\; (j = MP, FRN, CBI), \qquad |c_{14} + c_{24}| < |c_{44}|, \tag{13}$$

$$|c_{11} + c_{21}| > |c_{14} + c_{24}|, \qquad |c_{41}| < |c_{44}|. \tag{14}$$

The restrictions in (12) and (13) are implications of the model of Section 2. Those in (14) are not, because they compare MP with RP, whose relative size the model leaves free (Subsection 2.6).

Finally, the risk premium shock is required to correlate negatively with the change in the daily risk-appetite index of Bauer et al. (2023) over a two-day window centered on the announcement day, $\Delta BBM_t = BBM_{t+1} - BBM_{t-1}$:[6]

$$\operatorname{corr}\left(\varepsilon_t^{RP},\ \Delta BBM_t\right) < 0. \tag{15}$$

A positive ΔBBM_t is a risk-on move, and the risk premium shock is normalized as a risk-off shock. This is the only restriction that uses information from outside the announcement window, and Section 5 reports the decomposition without it.

Table 2 summarizes the full set of identifying restrictions in all three systems. The JKCS restrictions that neither JK nor CS impose concern the FRN shock, the split of the two-year yield into RES and FIT, the maturity ordering of MP, and the BBM correlation; none of them is needed to separate CBI from RP, which the orderings of CS already do.

[6]The index is available at daily frequency until 27 May 2022 and is constructed from a wide range of risk indicators drawn from equity, fixed-income, credit, and FX markets. Bauer et al. (2023) publish it as daily changes, and BBM_t is the cumulated index. The correlation is computed over the 335 windows for which ΔBBM_t is defined.

Table 2: Identifying restrictions: the JK, CS, and JKCS systems

	ε_t^{MP}	ε_t^{FRN}	ε_t^{CBI}	ε_t^{RP}	Magnitude and other restrictions
Panel A: JK system					
RES	+	*free*	*free*	n.a.	$b_{2Y,CBI} > 0$
FIT	*free*	+	*free*	n.a.	$\lvert b_{\text{FIT},MP}\rvert < b_{\text{RES},MP}$
ΔS&P500	−	−	+	n.a.	$\lvert b_{\text{RES},FRN}\rvert < b_{\text{FIT},FRN}$
Panel B: CS system					
ΔS&P500	−	n.a.	+	−	$\lvert b_{2Y,CBI}\rvert > \lvert b_{10Y,CBI}\rvert$ $\lvert b_{2Y,RP}\rvert < \lvert b_{10Y,RP}\rvert$
ΔR_{2Y}	+	n.a.	+	−	$\lvert b_{2Y,MP}\rvert > \lvert b_{2Y,RP}\rvert$ $\lvert b_{10Y,MP}\rvert < \lvert b_{10Y,RP}\rvert$
ΔR_{10Y}	+	n.a.	+	−	
Panel C: JKCS system					
RES	+	*free*	$+^{\S}$	$-^{\S}$	$\lvert b_{\text{FIT},MP}\rvert < b_{\text{RES},MP}$ $\lvert b_{\text{RES},FRN}\rvert < b_{\text{FIT},FRN}$
FIT	*free*	+	*free*	*free*	$\lvert b_{\text{RES},FRN}\rvert < b^{\S}_{\text{RES},MP}$ $\lvert b_{\text{FIT},MP}\rvert < b^{\S}_{\text{FIT},FRN}$
$\Delta R_{2Y} = \text{RES} + \text{FIT}$	+	$+^{\S}$	+	−	$\lvert b_{\text{FIT},CBI}\rvert < b^{\S}_{\text{RES},CBI}$ $\lvert b_{\text{FIT},RP}\rvert < \lvert b_{\text{RES},RP}\rvert^{\S}$
ΔS&P500	−	−	+	−	$\lvert b_{2Y,MP}\rvert > \lvert b_{10Y,MP}\rvert^{\S}$ $\lvert b_{2Y,FRN}\rvert > \lvert b_{10Y,FRN}\rvert^{\S}$
ΔR_{10Y}	+	$+^{\S}$	+	−	$\lvert b_{2Y,CBI}\rvert > \lvert b_{10Y,CBI}\rvert$ $\lvert b_{2Y,RP}\rvert < \lvert b_{10Y,RP}\rvert$
					$\lvert b_{2Y,MP}\rvert > \lvert b_{2Y,RP}\rvert$ $\lvert b_{10Y,MP}\rvert < \lvert b_{10Y,RP}\rvert$
					$\operatorname{corr}(\varepsilon_t^{RP}, \Delta\text{BBM}_t) < 0^{\S}$

Notes: $b_{m,j}$ is the impact response of observable m to shock j. In Panel C, $b_{2Y,j} \equiv b_{\text{RES},j} + b_{\text{FIT},j}$, so the two-year conditions are internal inequalities on the impact matrix rather than restrictions on a fifth observable. "n.a." means the shock is not identified in that system; "free" means the entry carries no sign restriction. In each system one sign per column is a normalization that labels the shock. JK also sharpen inference using heteroskedasticity between FOMC and non-FOMC events, which is not implemented here.
§ Imposed here but in neither 3-by-3 system.

Estimation of all three systems follows the rejection method of Rubio-Ramírez et al. (2010) and gives an identified set of impact matrices; point estimates are at the rotation of Fry and Pagan (2011). Appendix 8.3 states the algorithm.

3.4 Making the three systems comparable

The three systems are estimated on the same data and inputs, so the differences between their shocks come from identification: each three-shock system omits a shock, and its restrictions differ from those of JKCS. Appendix 8.6 separates the two by re-estimating JK and CS under the restrictions of JKCS.

1. Sample. All three systems are estimated on the event sample of Section 4, which uses scheduled and unscheduled announcements together with the press conferences.

2. JK modifications. Jarociński and Karadi (2026) build the policy surprise from the first principal component of the four nearest eurodollar futures. I use the two-year Treasury surprise instead, so that RES + FIT = ΔR_{2Y} and RES, FIT and ΔS&P500 are numerically the same series in all three systems. The cost of the choice is small: with the eurodollar measure in place of the two-year surprise, the JK MP and CBI shocks correlate with those used here at 0.97 or above. The heteroskedasticity step is not used (Subsection 3.1); the remaining restrictions are those of Jarociński and Karadi (2026), and the predictor set in (2) is the seventeen-variable set of Table 4.

3. CS as published. The Cieslak–Schrimpf system enters with all restrictions imposed in the published paper.

4 Data

The paper uses several types of data. First, high-frequency surprises around FOMC communication events and public information released before each event identify the shocks. Second, daily financial series and monthly macroeconomic indicators are the outcome variables for the effects of the shocks outside the announcement windows.

4.1 Announcement windows

FOMC surprises come from the U.S. Monetary Policy Event-Study Database of Acosta et al. (2025), which starts in February 1994. Earlier announcements, and one in October 2019, come from the event file of Bauer and Swanson (2023b). In the database, a statement surprise is the change in an asset price over a 30-minute window, from 10 minutes before the release to 20 minutes after. A press-conference surprise is the change over a 70-minute window, from 10 minutes before the Chair begins to 60 minutes after. The sample runs from the first announcement with a two-year yield surprise in the event file, in January 1991, to March 2025. It uses scheduled and unscheduled statements and press conferences. Some of the robustness checks additionally include the minutes releases. Unlike Jarociński and Karadi (2026), the sample has no Chair speeches or congressional testimonies, which the database does not cover. The sample excludes all events of March 2020. Table 3 reports the composition of the sample: 382 windows on 296 dates.

Table 3: The event sample

	Count
Announcement dates	296
Scheduled meetings	271
Unscheduled meetings	25
With a press conference	81
Statement windows	301
Press-conference windows	81
Windows used in the identification	382

Notes. January 1991 to March 2025, without the seven windows of March 2020. A press conference shares the date of the statement it follows, and five dates in 1991 have two statement windows; the 382 windows therefore fall on 296 dates. Windows with a missing predictor are dropped. The same event sample enters all three identification systems.

Four observables enter the JKCS system: the Bauer–Swanson residual (RES) and fitted value (FIT), constructed from the two-year Treasury yield surprise and the seventeen predictors listed in Table 4; the S&P500 surprise ΔS&P500 from the cash index; and the ten-year Treasury yield surprise ΔR_{10Y}. The JK system uses the first three; the CS system uses the two-year surprise $\Delta R_{2Y} = \text{RES} + \text{FIT}$, the ten-year surprise and the S&P500 surprise.

4.2 Predictors of the policy surprise

The predictor set Z_t in

$$\text{MPS}_t = \underbrace{\hat{\beta}' Z_t}_{\text{FIT}_t} + \underbrace{\widehat{\varepsilon}_t}_{\text{RES}_t},$$

follows Jarociński and Karadi (2026): it is the union of the regressors of Bauer and Swanson (2023a), Bauer and Swanson (2023b) and Swanson (2024); see Table 4 for definitions. As in Jarociński and Karadi (2026), the regression is estimated by OLS on all windows of Table 3; a press conference takes the predictors, and so the fitted value, of the statement it follows.

Table 4: Predictors of the policy surprise

Predictor	Definition	Source
Panel A. Bauer and Swanson (2023b)		
Payroll growth	Twelve-month log change in nonfarm payrolls, to the latest release	PAYEMS, FRED
Treasury skewness	Implied skewness of the ten-year Treasury yield from options on ten-year note futures	Bauer and Chernov (2024) series
Stock returns	Three-month change in the log S&P500	Bloomberg
Yield-curve slope	Three-month change in the second principal component of one- to ten-year zero-coupon Treasury yields	Federal Reserve Board (Gürkaynak et al. (2007))
Commodity prices	Three-month change in the log Bloomberg Commodity Index	Bloomberg
Panel B. Bauer and Swanson (2023a)		
Payrolls surprise	Released value minus the median forecast for that release	MMS survey, Haver Analytics
Unemployment surprise	As above, for the unemployment rate	As above
Real GDP surprise	As above, for real GDP; advance, second and third releases stacked by release date	As above
Core CPI surprise	As above, for core CPI	As above
Core CPI forecast	Median forecast for the core CPI release of the announcement month	As above
Core CPI acceleration	$200\,(\ln P_{m-2} - 2\ln P_{m-8} + \ln P_{m-14})$, with P the core CPI index	CPILFESL, FRED
Business cycle index	Brave–Butters–Kelley cyclical component, month $m-1$	BBKMCY, FRED
Panel C. Swanson (2024)		
Two-year yield	Three-month change in the two-year zero-coupon Treasury yield	Federal Reserve Board (Gürkaynak et al. (2007))
Ten-year yield	Three-month change in the ten-year zero-coupon Treasury yield	As above
Baa spread	Three-month change in the Moody's Baa yield less the ten-year Treasury yield	BAA10Y, FRED
Shadow rate	Three-month change in the Wu–Xia shadow federal funds rate	Wu and Xia (2016) series
Financial conditions	Four-week change in the Adjusted National Financial Conditions Index	ANFCI, Chicago Fed

Notes. Several regressors appear in more than one of the three papers. For the implications of the choice of regressors, see Laumer (2026).

4.3 Other data

The local projections in Section 6.1 use 14 daily series: the two- and ten-year breakeven inflation rates, the five-year investment-grade and high-yield CDX total-return indices, and the S&P500, all from Bloomberg; the VIX and the ICE BofA BBB corporate spread; the one-, two- and ten-year yields of Adrian et al. (2013), with the decomposition of the ten-year yield into an expectations component and a term premium, from the Federal Reserve Bank of New York; the risk appetite index of Bauer et al. (2023); and the risk aversion index of Bekaert et al. (2022), the last two from the authors' data pages. With one exception these

series play no role in the identification. The exception is the risk appetite index, which enters through the external correlation restriction (15); its response to the RP shock around the announcement is therefore partly imposed. The risk aversion index replaces it in a robustness check of Section 5 and enters no restriction in the baseline.

The VARs in Sections 6.2–6.4 use ten monthly variables: the one- and ten-year Treasury yields, the log S&P500, the log PCE price index and log housing starts, all from FRED-MD; the excess bond premium of Gilchrist and Zakrajšek (2012), from the Federal Reserve Board's updated series; and the logs of monthly interpolations of real GDP and of total, residential and nonresidential real private fixed investment.

5 What the four shocks explain within the window

I now turn to how much of each surprise the four shocks explain, and to what changes when one of them is left out, as in the JK and CS systems. All three systems are estimated on the same event sample, so no difference reported below comes from the sample.

Let $u_t = B\varepsilon_t$ be the vector of window surprises, and let b_{mj} be the response of observable m to a one-standard-deviation shock j. The shocks are orthogonal and have unit variance, so the variance of observable m across all windows is $\sum_k b_{mk}^2$, and the share explained by shock j is

$$s_{mj} = \frac{b_{mj}^2}{\sum_k b_{mk}^2}.$$

The shares sum to one across shocks by construction, and they describe the sample as a whole, not any single announcement. Because the system has as many observables as shocks, the four shocks explain all the variation in the observables; the identifying restrictions decide only how this variation is split among them.

5.1 Omitting the Fed response to news shock

Table 5 compares the variance shares in the four-shock system with those in the three-shock CS system, for the three CS observables: the S&P500, the two-year yield and the ten-year yield. The four-shock system itself has four observables: RES, FIT, the S&P500 and the ten-year yield. The two-year yield is not a fifth one; it equals RES+FIT, so its response to each shock is the sum of the RES and FIT responses. The Δ column is the difference between the two systems. The last column is the percentage of rotations in the four-shock identified set that give the difference the same sign as the reported point estimate, with CS held at its Fry–Pagan rotation. A value near 50 means that the sign of the difference is not determined;

a value below 50 means that most rotations give the opposite sign.

In the four-shock system, each shock accounts for at least a fifth of the variance of one of the three surprises. The dominant shocks differ: the policy shock for the equity surprise, the policy and CBI shocks together for the two-year surprise, and the risk premium shock for the ten-year surprise, where it accounts for about 60% of the variance.

Adding the FRN shock mainly lowers the policy shock's share, as the model predicts. The FRN shock accounts for 11 to 21% of the variance of the three surprises, and the policy shock's share falls by 9 to 25 points, the largest decline in each surprise.

The information share rises for the equity surprise, falls for the two-year surprise and barely changes for the ten-year surprise; the risk premium share falls by less than two points in all three.

Omitting the FRN shock therefore mainly distorts the policy shock: in the CS system, the policy shock absorbs the Fed response to news. This is also the most robust result in the table. The decline in the policy shock's share holds in majority of the rotations in the identified set (77 - 88%), while every other change holds in at most 63%, close to the 50% at which its direction would be undetermined.

Table 5: Announcement-window variance decomposition: omitting the Fed response to news shock

Shock	CS (3 shocks)	JKCS (4 shocks)	Δ (pp)	sign (%)
$\Delta S\&P500$, $\sigma = 0.60\%$				
MP	92.1 [62.7, 99.3]	66.8 [35.2, 95.5]	−25.3	88
FRN	n.a.	21.3 [0.2, 53.5]	+21.3	n.a.
CBI	6.0 [0.0, 36.7]	10.3 [0.0, 32.2]	+4.3	43
RP	1.9 [0.0, 7.9]	1.7 [0.0, 9.9]	−0.2	62
ΔR_{2Y}, $\sigma = 5.26$ *bp*				
MP	53.1 [31.2, 85.8]	40.6 [18.7, 70.6]	−12.5	77
FRN	n.a.	20.0 [1.8, 29.0]	+20.0	n.a.
CBI	42.7 [12.5, 65.3]	36.3 [13.2, 61.3]	−6.4	62
RP	4.2 [0.1, 16.6]	3.1 [0.1, 15.6]	−1.1	50
ΔR_{10Y}, $\sigma = 4.25$ *bp*				
MP	35.2 [21.6, 47.4]	26.4 [11.2, 43.2]	−8.8	78
FRN	n.a.	10.9 [0.1, 29.7]	+10.9	n.a.
CBI	3.7 [0.0, 18.5]	3.2 [0.0, 15.0]	−0.5	63
RP	61.0 [47.8, 75.5]	59.5 [43.2, 73.5]	−1.6	61

Notes. CS is the three-shock Cieslak–Schrimpf system as published; JKCS is the four-shock system mapped to the same three variables, where RES + FIT = ΔR_{2Y}. Both decompose the same $\hat{\Sigma}$ over the same 382 windows. Point shares correspond to Fry–Pagan rotations; brackets show 5–95% ranges of each identified set.

5.2 Omitting the risk premium shock

Table 6 does the same for the Jarociński–Karadi system, which has no risk premium shock. In the four-shock system, the risk premium shock accounts for less than 5% of the variance of RES, FIT, the equity surprise and the two-year yield. This does not make the shock unimportant: it mainly moves the ten-year yield, which the JK system does not observe and the table therefore does not show. Still, the four-shock system splits the variance among the other three shocks differently from JK.

In the model, FRN is the only shock in FIT, the predictable part of the policy surprise, because FIT depends only on information known before the announcement, while MP, CBI and RP are news at the announcement. The restrictions do not impose this, and the four-shock system attributes almost 90% of FIT to FRN, against about 80% in JK, mostly at the expense of the CBI shock. The largest changes are in the policy shock's share, by 14 to 28 points: it falls for RES and the two-year yield and rises for the equity surprise. Moving from

JK to the four-shock system therefore changes the shares of all three shared shocks, and by more than moving from CS does in Table 5: the largest change is 28 points here and 25 there. The changes also keep their sign more often across the identified set.

These differences cannot be read as the effect of the missing shock alone. The JK system is estimated with its published restrictions, so it differs from the baseline in its restrictions as well as in its shocks. Beyond the JK restrictions, the four-shock system requires the MP shock to move RES more than the FRN shock does, the FRN shock to move FIT more than the MP shock does, and the CBI shock to raise RES and to move it more than FIT. The same point applies, to a lesser degree, to the comparison with CS: unlike CS, the four-shock system requires the MP shock to move the two-year yield more than the ten-year yield, and the risk premium shock to correlate negatively with the risk appetite index. In Appendix 8.6, I re-estimate both three-shock systems under the restrictions of the four-shock baseline. The restrictions explain little of the gap between CS and JKCS, but most of the gap between JK and JKCS.

Of 10 million candidate rotations, 975 satisfy every restriction of the four-shock system. The sign restrictions alone remove all but about 1% of the candidates; the magnitude restrictions on RES and FIT and the maturity orderings remove most of the rest; the comparisons with the policy shock remove about one in nine of the rotations that remain, and the correlation with the risk appetite index removes none. When the restrictions are dropped one at a time, the set grows most without the requirement that the FRN shock move the two-year yield more than the ten-year yield: it becomes about three times larger.

Table 6: Announcement-window variance decomposition: omitting the risk premium shock

Shock	JK (3 shocks)	JKCS (4 shocks)	Δ (pp)	sign (%)
RES, $\sigma = 5.00$ *bp*				
MP	85.5 [33.3, 99.2]	59.7 [29.0, 86.0]	−25.8	95
FRN	0.0 [0.0, 7.4]	2.8 [0.0, 7.4]	+2.7	94
CBI	14.5 [0.1, 63.9]	34.9 [8.8, 63.0]	+20.4	90
RP	n.a.	2.7 [0.2, 15.4]	+2.7	n.a.
FIT, $\sigma = 1.62$ *bp*				
MP	3.0 [0.0, 21.1]	9.9 [0.0, 24.1]	+6.9	59
FRN	82.1 [19.0, 98.9]	87.9 [48.9, 97.9]	+5.8	59
CBI	14.9 [0.1, 77.4]	1.7 [0.0, 30.1]	−13.2	83
RP	n.a.	0.4 [0.0, 14.6]	+0.4	n.a.
Δ*S&P500*, $\sigma = 0.60\%$				
MP	53.2 [16.0, 93.7]	66.8 [35.2, 95.5]	+13.6	79
FRN	23.4 [0.2, 70.6]	21.3 [0.2, 53.5]	−2.1	64
CBI	23.5 [0.2, 61.9]	10.3 [0.0, 32.2]	−13.2	90
RP	n.a.	1.7 [0.0, 9.9]	+1.7	n.a.
ΔR_{2Y}, $\sigma = 5.26$ *bp*				
MP	68.2 [22.8, 96.9]	40.6 [18.7, 70.6]	−27.6	93
FRN	8.6 [0.1, 28.7]	20.0 [1.8, 29.0]	+11.4	73
CBI	23.1 [0.2, 65.1]	36.3 [13.2, 61.3]	+13.1	82
RP	n.a.	3.1 [0.1, 15.6]	+3.1	n.a.

Notes. JK is the three-shock Jarociński–Karadi system with sign and magnitude restrictions as published. $\Delta R_{2Y} = \text{RES} + \text{FIT}$ is not a separate observable, and ΔR_{10Y} is not in the table because the JK model does not include it among its observables. Both identification schemes decompose the same $\hat{\Sigma}$ over the same 382 windows. Point shares correspond to Fry–Pagan rotations; brackets show 5–95% ranges of each identified set.

5.3 What each three-shock system recovers

Next, I show how the shocks of each three-shock system relate to the four shocks of the joint system. The JK and CS observables are combinations of the JKCS observables, so each JK or CS shock is an exact combination of the four JKCS shocks, $\tilde{\varepsilon}_j = \sum_i \lambda_{ij}\varepsilon_i$, with weights that satisfy $\sum_i \lambda_{ij}^2 = 1$. Table 7 reports $100\lambda_{ij}^2$, the share of each JK or CS shock accounted for by each JKCS shock. The four entries in each column sum to 100. A column with one entry near 100 means that the shock is the same in both systems; otherwise, the shock mixes several JKCS shocks, in the proportions shown.

The column Σ, the row sum, is the R^2 from regressing a JKCS shock on the three shocks of the JK or CS system. Several results follow from the table. First, in every column the largest entry is on the row of the same name, so the shock labels broadly agree across systems; that entry is about three quarters or more. Second, about a quarter of what the CS system calls a monetary policy shock is the FRN shock. Third, a fifth of the JK CBI shock comes from the other three shocks, the largest part from the RP shock, so part of the information shock in the JK system is risk-premium variation. Finally, the CS shocks explain only 73% of the JKCS policy shock, and the JK shocks 90% of the JKCS information shock.

Table 7: How shocks are related across identifications

Panel A. JK system (3 shocks)				
JKCS shock	JK MP	JK FRN	JK CBI	Σ
MP	**91.3** [84.2]	**0.2** [2.8]	**8.4** [9.1]	99.9
FRN	**0.9** [1.5]	**96.2** [87.7]	**2.9** [6.5]	100.0
CBI	**7.5** [10.7]	**3.3** [4.2]	**79.7** [67.3]	90.5
RP	**0.3** [0.7]	**0.2** [1.6]	**9.1** [10.1]	9.6
Panel B. CS system (3 shocks)				
JKCS shock	CS MP	CS CBI	CS RP	Σ
MP	**73.4** [77.3]	**0.0** [2.4]	**0.0** [0.6]	73.4
FRN	**26.0** [18.9]	**1.3** [3.3]	**0.0** [1.5]	27.3
CBI	**0.6** [1.8]	**98.5** [90.0]	**0.2** [1.3]	99.3
RP	**0.0** [0.5]	**0.2** [1.2]	**99.8** [94.8]	100.0

Notes. Entries are the share of each JK and CS shock accounted for by each JKCS shock, in %; the four entries in a column sum to 100. Σ is the row sum, the share of each JKCS shock the three-shock system explains. Points are Fry–Pagan rotations, brackets the median over the JKCS identified set. All three systems are identified on the 382 windows without March 2020.

Table 8 shows the same in terms of correlations. The correlations between the versions of a shock in different systems range from 0.85 to 1.00. The lowest are for the CS policy shock, which the FRN shock contaminates, and for the JK information shock, which the RP shock contaminates. These are the two shocks whose series depend on the choice of system.

Table 8: Pairwise correlations of shocks

Shock	JK–JKCS	JK–CS	JKCS–CS
MP	0.96	0.85	0.86
CBI	0.89	0.89	0.99
FRN	0.98	-	-
RP	-	-	0.99

Notes. Correlations on the same 382 windows, with each system at its Fry–Pagan rotation. "-" marks a shock absent from one of the two systems.

One property that none of the three systems imposes can be used to check them: a structural shock should not be predictable from its own past. The FRN shock is the exception, because it drives the part of the surprise that is predictable from public information and so should be autocorrelated. Table 9 reports the first-order autocorrelation across consecutive announcement days. The autocorrelation of the policy shock is not significantly different from zero in any of the three systems. The autocorrelation of the FRN shock is significant at the 1% level in both systems that identify it, as its definition implies. The autocorrelation of the CBI shock is not significant at the 10% level in any of the three systems. The risk premium shock has a small negative autocorrelation in JKCS and CS, significant at the 5% level.

Table 9: First-order autocorrelation of the identified shocks

Shock	JK	JKCS	CS
MP	-0.04	0.02	-0.04
CBI	0.07	-0.01	0.00
FRN	0.32^{***}	0.34^{***}	-
RP	-	-0.12^{**}	-0.12^{**}

Notes. Autocorrelation is measured across consecutive FOMC announcements (296 days in total). Shocks are at the Fry–Pagan rotation of each system, with same-day windows summed, so the lag is one announcement day. Stars mark rejection of the null of zero first-order autocorrelation at the 1, 5 and 10% levels. "-": a shock absent from that system.

5.4 The shocks and the Swanson (2021) factors

The three factors of Swanson (2021) are an alternative decomposition of the news in each FOMC announcement: a federal funds rate (FFR), a forward guidance (FG) and an asset-purchase (LSAP) factor. Swanson's factors use none of the restrictions of this paper, so comparing them with the four identified shocks is a check on the labels of the shocks. Each

column of Table 10 reports a regression of one of Swanson's factors on the four identified shocks. Same-day statement and press-conference windows are summed into one observation per announcement date. The second and third groups of columns restrict the sample to scheduled meetings and to 2009–2015, the years of the most active asset purchases.

The FFR factor mainly mixes the policy and FRN shocks, the FG factor mixes all four shocks, and the LSAP factor is mostly the risk premium shock. The policy and FRN shocks enter the FFR factor together because they move every asset price proportionally and differ only in whether they were predictable from pre-announcement data. Since Swanson's factors use no pre-announcement information, they cannot separate the two. A risk premium shock that lowers the ten-year yield also lowers the FG factor, as dovish forward guidance would, so part of measured forward guidance is premium variation. The LSAP factor, in contrast, reflects mainly one shock: in all three samples the risk premium coefficient is between 0.35 and 0.40 and no other shock comes close. Over 2009–2015 the four shocks explain nearly all of its variation. Thus, in those years the LSAP factor and the risk premium shock capture largely the same variation.

The risk premium shock also enters the FFR factor, but mostly through the unscheduled announcements: without them, its coefficient is no longer significant at the 10% level. The fit of the FFR regression falls by more than half in both subsamples, and the four shocks leave most of the factor's variation unexplained. All four coefficients in the FG column keep their signs and stay significant at the 5% level in both subsamples.

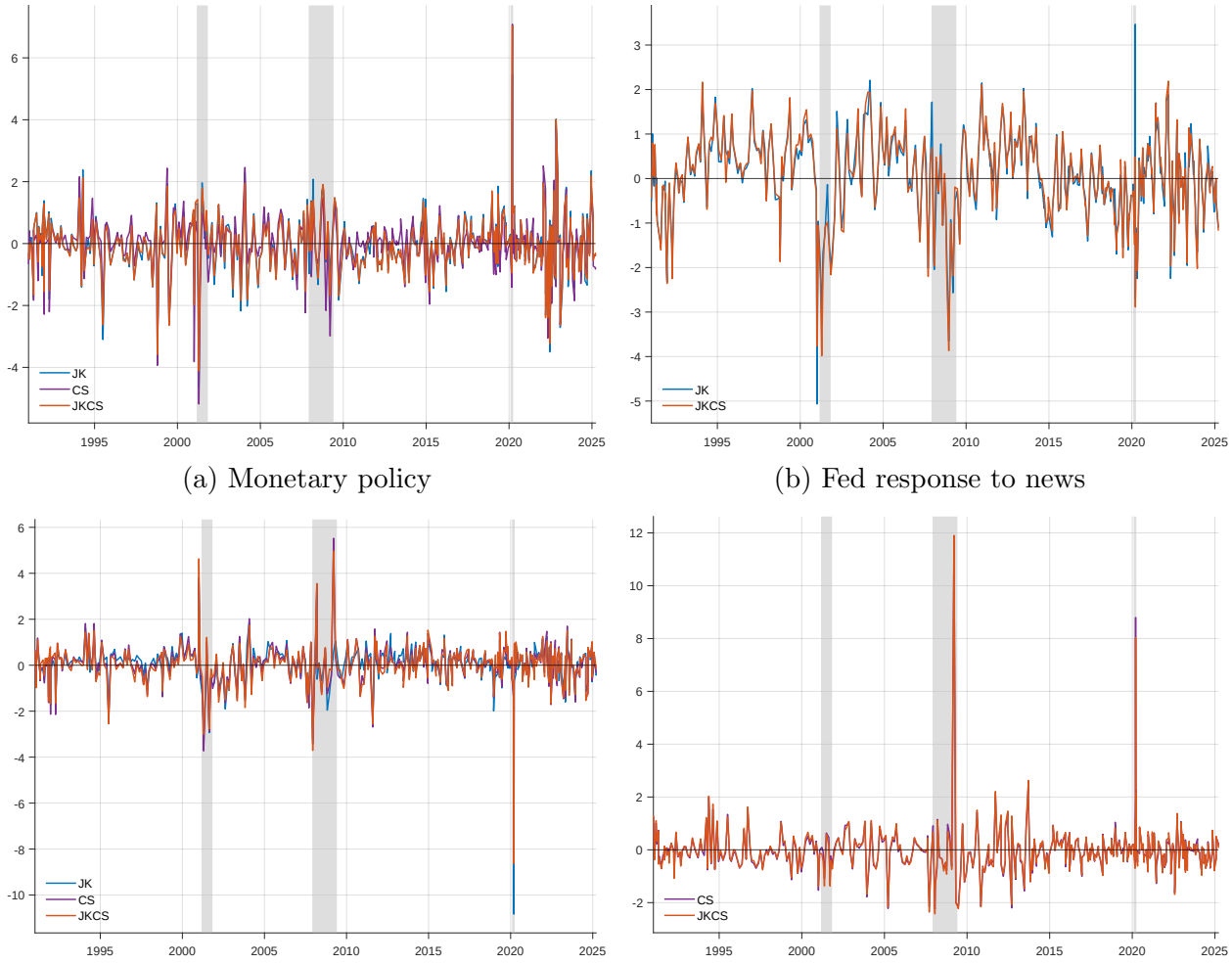
Table 10: Swanson (2021) factors regressed on the four JKCS shocks

	All meetings			Scheduled meetings			LSAP period (2009–2015)		
	FFR	FG	LSAP	FFR	FG	LSAP	FFR	FG	LSAP
MP	0.329^{***}	0.382^{***}	-0.028	0.187^{***}	0.526^{***}	-0.012	-0.018	0.663^{***}	-0.008
FRN	0.340^{***}	0.279^{***}	0.028	0.144^{***}	0.442^{***}	0.050^{*}	-0.000	0.519^{***}	-0.099^{**}
CBI	0.131^{**}	0.405^{***}	0.159^{***}	0.089^{**}	0.443^{***}	0.174^{***}	0.000	0.493^{***}	0.100^{*}
RP	0.121^{**}	-0.331^{***}	0.348^{***}	0.049	-0.276^{***}	0.351^{***}	-0.032^{**}	-0.162^{***}	0.396^{***}
N		239			223			55	
R^2	0.437	0.639	0.631	0.173	0.740	0.664	0.204	0.835	0.907

Notes. Each column regresses a Swanson (2021) factor on the four JKCS shocks, evaluated at the Fry–Pagan rotation, with same-day statement and press-conference windows summed to one observation per announcement date. Standard errors are HC1 heteroskedasticity-robust. Samples: all FOMC meetings, scheduled meetings only, and the LSAP period January 2009 – December 2015; the factors end in June 2019. *** ** *: significance at the 1%, 5%, and 10% levels.

5.5 The shock series

Figures 2a–2d plot each shock as identified by each system that contains it. Where a shock appears in more than one system, the series track each other closely over three decades, in recessions and outside them, as the high correlations imply. The two risk premium series are almost identical at every date, including the largest realizations (Figure 11 in the Appendix compares their distributions and plots one against the other). The monetary policy shock is the exception: the JK and JKCS series are hard to tell apart, but the CS series departs from both at the dates where the Fed response to news is largest.



(a) Monetary policy

(b) Fed response to news

(c) Central bank information

(d) Risk premium

Figure 2: Shock series across identification systems

Notes. Each line is a shock at the Fry–Pagan rotation of its own system, in standard deviations (each series has unit variance over the $N = 382$ announcement windows, 1991–2025). Grey areas are NBER recessions.

5.6 One announcement, three systems

The comparisons above show how the three systems differ on average. To see what this means for a single announcement, consider 18 September 2007, when the FOMC cut the target rate by 50 basis points. It was the first cut of the easing cycle, after financial markets had started to experience liquidity shortages. Figure 3 decomposes the window response of the two-year yield, the ten-year yield and the S&P500 into contributions from each shock, one row per system. In the 30-minute window, the two-year yield fell and the S&P500 rose, consistent with an easing surprise, but the ten-year yield rose slightly.

In the four-shock system (last row), the policy, Fed response to news and information shocks all push both yields down. The announcement combined an unexpected easing of the policy rate, bad news about the economy, and a stronger response to that news than the market had expected. Although all three push rates down, the ten-year yield rose. Under the maturity restrictions, only the risk premium shock moves the ten-year yield more than the two-year yield, so the decomposition attributes the rise in the ten-year yield to it.

Each three-shock system misses part of this picture. The JK system of Jarociński and Karadi (2026) has no risk premium shock and does not observe the ten-year yield. The CS system has no Fed response to news shock, so its policy shock absorbs the Fed response to news: it attributes to the policy shock about as much of the equity rise as the four-shock system attributes to the MP and FRN shocks together. The estimated policy shock therefore depends on which other shocks the system allows.

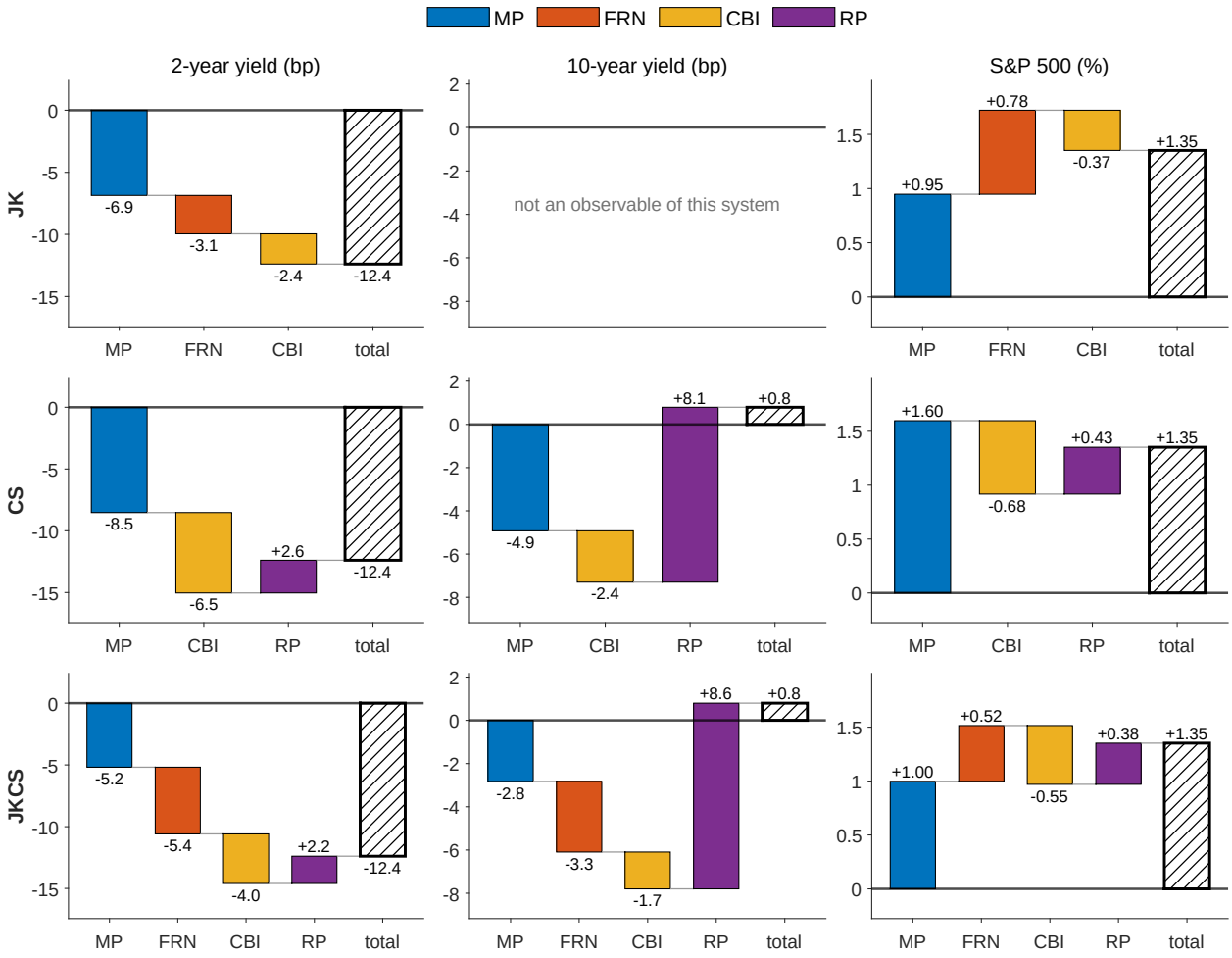


Figure 3: Decomposition of the announcement surprise, 18 September 2007

Notes. Each panel decomposes the surprise in one observable under one system, evaluated at that system's Fry–Pagan rotation. Bars are read left to right: each begins where the previous one ends. The hatched bar is the total actual change.

5.7 The full sample

Figures 4–6 report the decomposition at every announcement in the sample. Each bar splits one window's surprise in the two-year yield, the S&P500 or the ten-year yield into the contributions of the four shocks at the Fry–Pagan rotation; the four contributions add up to the observed change.

Throughout the sample, the two-year yield surprise comes from the policy, Fed response to news and information shocks. The risk premium shock adds little, and only in short episodes. The equity surprise is shared between the policy and Fed response to news shocks; the information and risk premium contributions are small at most announcements.

The risk premium shock matters most for the ten-year yield, where its contribution

exceeds that of any other shock. Part of this is built in, through two restrictions. The first, which the model implies, requires the risk premium shock to move the ten-year yield more than the two-year yield. The second, which the model does not imply, requires it to move the ten-year yield more than the policy shock does. No restriction compares it with the information and Fed response to news shocks, and none restricts the size of its contribution or its timing. Dropping the second restriction leaves its share of ten-year variance unchanged. Dropping every restriction on the risk premium shock makes the 5–95% band on its ten-year share about two and a half times as wide (Subsection 5.9).

Whether a contribution is positive or negative depends only on the sign of the shock in that window, because the response of each observable to one unit of a shock is the same at every date. A positive risk premium shock lowers stock prices and both yields, the ten-year yield the most; a negative one raises them. In all but two of the 28 windows where the risk premium shock exceeds 1.5 standard deviations in absolute value, it is positive when the ten-year yield falls and negative when it rises. The positive ones include the largest of all, the expansion of asset purchases on 18 March 2009, as well as the maturity extension program of 21 September 2011 and the decision not to taper of 18 September 2013. The negative ones include the rate cuts of 18 September 2007 and 22 January 2008, and the start of the second and third purchase programs on 3 November 2010 and 13 September 2012.

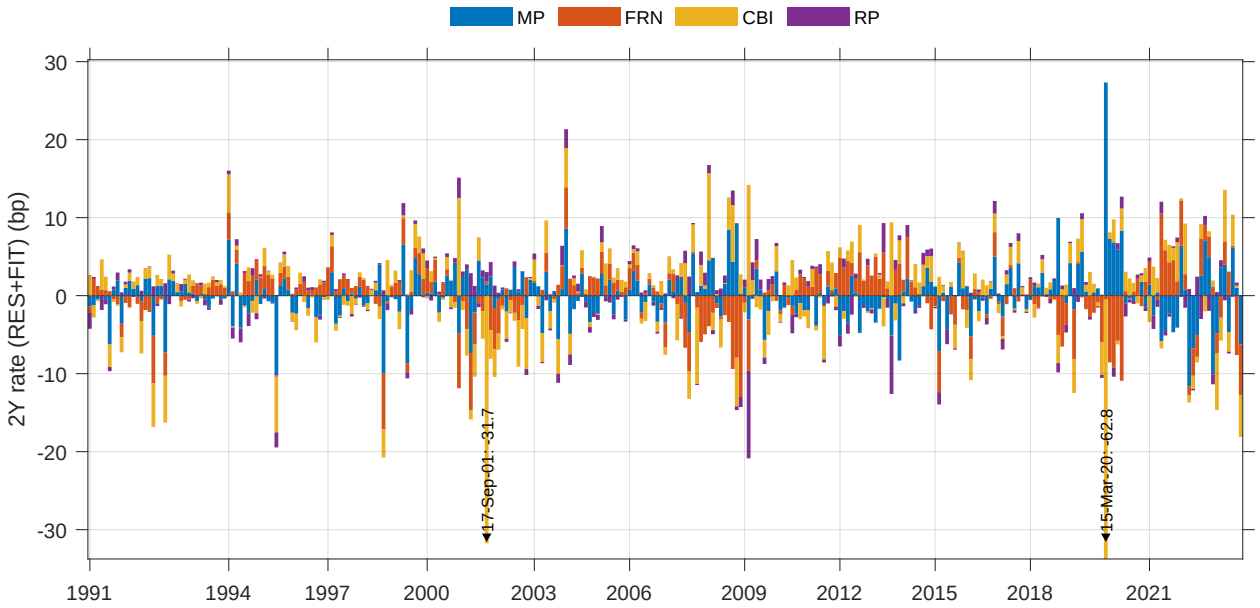


Figure 4: Contributions of the four shocks to the policy surprise, RES + FIT

Notes. Contributions are shown at every announcement, at the Fry–Pagan rotation. The four contributions sum to the observed change. Positive and negative contributions are drawn upward and downward from zero, so the observed change is the height of a bar minus its depth.

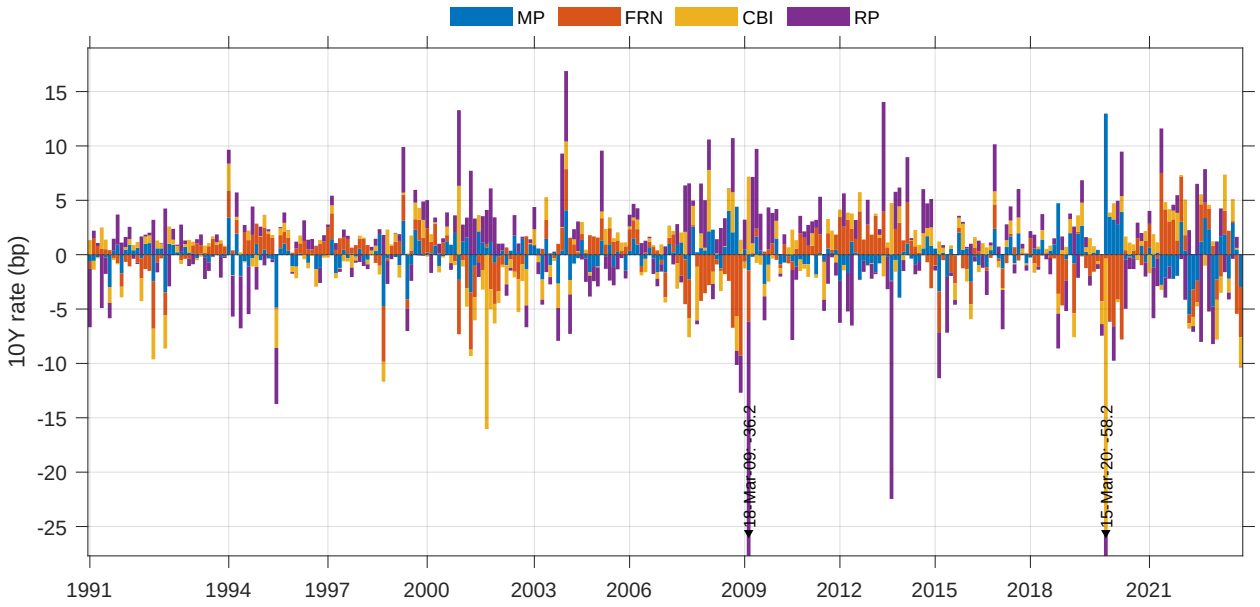


Figure 5: Contributions of the four shocks to the ten-year yield surprise

Notes. Layout as in Figure 4.

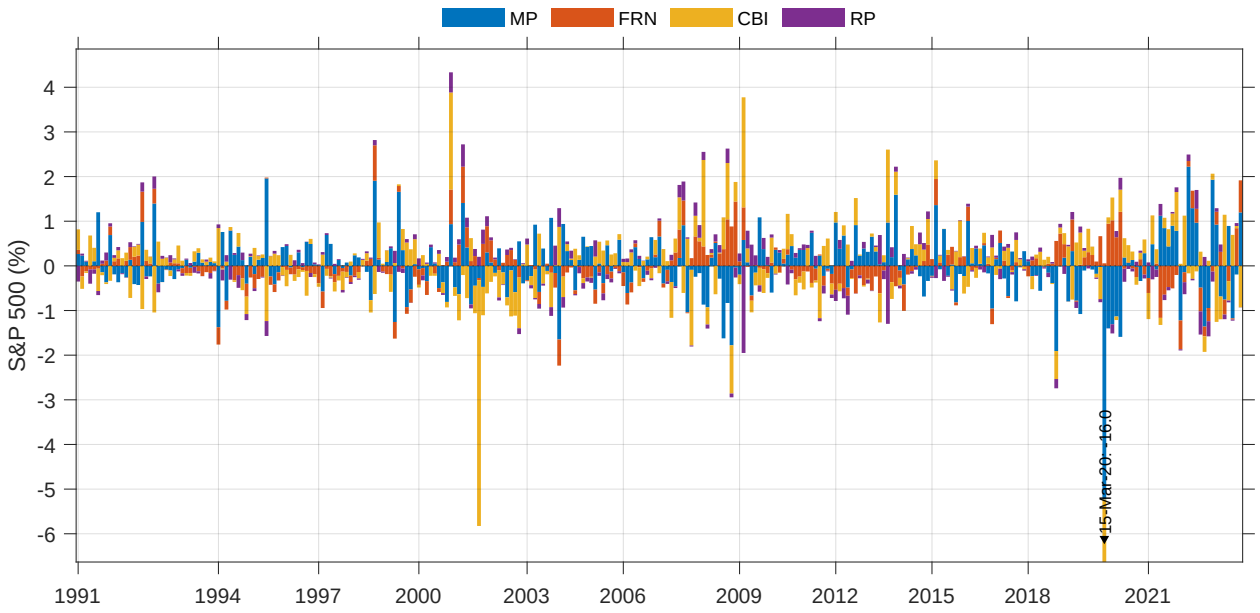


Figure 6: Contributions of the four shocks to the S&P500 surprise

Notes. Layout as in Figure 4.

5.8 Selected announcements

Table 11 reports the decomposition for five FOMC announcements: the unscheduled 75 basis point cut of 22 January 2008, the expansion of asset purchases of 18 March 2009,

the taper communication of 19 June 2013, the decision not to taper of 18 September 2013, and the 75 basis point increase of 15 June 2022. For each window, it reports the observed change in the ten-year yield and in the S&P500 and each shock's contribution, in the same units. The contributions add up to the observed change at every rotation, and the brackets give the $[5, 95]$ range of each contribution across rotations. Three of the announcements concerned the balance sheet and two were rate decisions. The identification does not use that distinction: it sees only the window surprises and the restrictions. The question is whether the decomposition separates the two kinds of announcement anyway.

On the three balance-sheet announcements, 18 March 2009, 19 June 2013 and 18 September 2013, the risk premium shock moves the ten-year yield. On 18 March 2009 the yield fell by 43 basis points, and the decomposition attributes 39 basis points of the decline to the risk premium shock. On 19 June 2013 the yield rose and on 18 September 2013 it fell, in both the statement and the press-conference windows; in all four windows the risk premium shock is the largest contributor, and on all three dates its $[5, 95]$ range excludes zero. For 18 March 2009 this agrees with the event-study evidence on QE1, which attributes most of the decline in the ten-year yield around the purchase announcements to the term premium (Gagnon et al., 2011).

On the rate decision of 15 June 2022, the risk premium shock contributes less than one basis point to the ten-year yield in either window, and the decline in the ten-year yield during the press conference is mostly the policy shock. On 22 January 2008, in contrast, the ten-year yield rose by about 9 basis points, and the decomposition attributes 8 of them to the risk premium shock, with a $[5, 95]$ range that excludes zero.

Nothing in the identification distinguishes a balance-sheet announcement from a rate decision, so the restrictions do not impose this pattern. A system without the risk premium shock has to attribute the long-end movement on the balance-sheet dates to the policy or information shock, which is the misattribution measured on average earlier in this section.

Table 11: Famous announcements: observed window changes and each shock's contribution.

		Observed	ε^{MP}	ε^{FRN}	ε^{CBI}	ε^{RP}
22-Jan-2008 (unscheduled): *Unscheduled intermeeting 75 bp cut*						
	ΔR_{10Y} (bp)	+8.6	+3.3 [+1, +5]	−2.7 [−3, −0]	−0.0 [−1, +1]	+8.1 [+5, +10]
	ΔS&P (%)	−0.0	−0.7 [−0.9, −0.2]	+0.5 [+0.0, +0.8]	−0.0 [−0.2, +0.2]	+0.2 [+0.0, +0.5]
18-Mar-2009: *QE1 expansion announced*						
	ΔR_{10Y} (bp)	−43.1	−4.7 [−14, +1]	−3.2 [−12, +1]	+4.1 [+0, +6]	−39.4 [−43, −27]
	ΔS&P (%)	+1.8	+1.0 [−0.3, +3.0]	+0.6 [−0.3, +2.0]	+1.0 [+0.0, +1.7]	−0.9 [−1.9, −0.0]
19-Jun-2013: *Taper communication*						
stmt	ΔR_{10Y} (bp)	+6.5	−1.1 [−2, +1]	+2.2 [+0, +4]	−0.1 [−1, +1]	+5.6 [+3, +7]
	ΔS&P (%)	−0.1	+0.3 [−0.1, +0.4]	−0.4 [−0.6, −0.0]	−0.0 [−0.1, +0.1]	+0.1 [+0.0, +0.3]
press	ΔR_{10Y} (bp)	+5.8	−0.1 [−1, +2]	+2.7 [+0, +5]	−0.8 [−2, −0]	+4.1 [+1, +6]
	ΔS&P (%)	−0.6	+0.0 [−0.5, +0.3]	−0.5 [−0.9, −0.0]	−0.2 [−0.4, −0.0]	+0.1 [+0.0, +0.2]
18-Sep-2013: *No-taper surprise*						
stmt	ΔR_{10Y} (bp)	−10.7	−3.2 [−6, −1]	−0.8 [−3, +0]	+1.2 [+0, +2]	−8.0 [−9, −5]
	ΔS&P (%)	+1.0	+0.7 [+0.2, +1.2]	+0.1 [−0.1, +0.6]	+0.3 [+0.0, +0.6]	−0.2 [−0.3, −0.0]
press	ΔR_{10Y} (bp)	−7.1	−0.8 [−2, +0]	−0.2 [−2, +0]	+0.7 [+0, +1]	−6.8 [−7, −5]
	ΔS&P (%)	+0.2	+0.2 [−0.1, +0.5]	+0.0 [−0.1, +0.2]	+0.2 [+0.0, +0.3]	−0.2 [−0.3, −0.0]
15-Jun-2022: *75 bp hike, first since 1994*						
stmt	ΔR_{10Y} (bp)	+1.9	+1.2 [+0, +2]	+0.4 [−0, +1]	+0.8 [+0, +2]	−0.4 [−1, +1]
	ΔS&P (%)	−0.1	−0.3 [−0.4, −0.1]	−0.1 [−0.1, +0.0]	+0.2 [+0.0, +0.3]	−0.0 [−0.0, +0.0]
press	ΔR_{10Y} (bp)	−9.6	−7.0 [−9, −4]	−0.9 [−2, +0]	−1.3 [−4, −0]	−0.4 [−3, +1]
	ΔS&P (%)	+1.4	+1.6 [+1.1, +1.8]	+0.2 [−0.1, +0.4]	−0.3 [−0.4, −0.0]	−0.0 [−0.1, +0.0]

Contributions are at the Fry–Pagan rotation and sum to the observed change; brackets give their [5,95] range across accepted rotations. Statement (stmt) and press-conference (press) windows are shown separately. Changes are net of the sample means of the window changes, -0.37 bp for the ten-year yield and +0.03% for the S&P500. A cut smaller than its predictable component is a positive policy shock.

5.9 Robustness

Table 12 reports the variance decomposition of the four-shock system in seven configurations: the baseline and six variants, each changing one element of the identification or of the event set. Panel A reports the risk premium share of each observable and Panel B the full decomposition of the ten-year yield. Rows (2) and (3) concern the external restriction, the negative correlation of the risk premium shock with an index of risk appetite: row (2) replaces the Bauer et al. (2023) index with the Bekaert et al. (2022) risk-aversion index, and row (3) drops the restriction. Row (4) drops the requirement that the risk premium shock move the ten-year yield more than the policy shock does. Rows (5) to (7) change the event set: they remove the unscheduled announcements, keep only the statement windows, and add the minutes releases.

The decomposition changes little across the seven configurations. The risk premium shock is the largest contributor to ten-year variance in all of them, with a share of 60 to 70%, more than twice that of any other shock. Replacing the index used for the external restriction leaves every point estimate unchanged. Likewise, row (4) reproduces the baseline point estimates, and at the center of the identified set this restriction is nearly redundant. Rows (1), (2) and (4) use the same event set and therefore the same reduced-form covariance matrix. The restrictions that differ between them do not bind at the Fry–Pagan rotation, so the point estimates coincide and only the bounds of the identified set move slightly.

Appendix 8.5 leaves the fourth column of B free of any sign, magnitude, or external restriction. The shock that comes out still lowers the unpredictable part of the policy surprise, and the other three shocks are almost unchanged. The restrictions buy precision: without them, the 5–95% band on the risk premium share of ten-year variance is about two and a half times as wide, 78 percentage points instead of 30.

Table 12: Robustness of the four-shock decomposition

Configuration	RES	FIT	ΔS&P500	ΔR_{10Y}
Panel A. Share of each variable attributed to the risk premium shock (%)				
(1) Baseline (BBM ext. rest.)	2.7 [0.2, 15.4]	0.4 [0.0, 14.6]	1.7 [0.0, 9.9]	59.5 [43.2, 73.5]
(2) BEX instead of BBM	2.7 [0.2, 15.4]	0.4 [0.0, 14.6]	1.7 [0.0, 9.9]	59.5 [43.2, 73.5]
(3) No ext. rest.	2.7 [0.2, 15.4]	0.4 [0.0, 14.6]	1.7 [0.0, 9.9]	59.5 [43.2, 73.5]
(4) − RP moves R_{10Y} more than MP	2.7 [0.2, 15.0]	0.4 [0.0, 14.4]	1.7 [0.0, 14.7]	59.5 [37.6, 73.3]
(5) Scheduled meetings only	4.4 [0.2, 11.6]	3.8 [0.0, 17.7]	0.7 [0.0, 9.9]	69.7 [47.8, 76.7]
(6) Statement windows only	5.3 [0.2, 15.7]	0.2 [0.0, 13.9]	1.1 [0.0, 10.5]	65.5 [43.0, 73.5]
(7) Minutes releases added	5.3 [0.2, 15.4]	0.2 [0.0, 14.5]	1.0 [0.0, 9.7]	65.3 [45.7, 74.3]
Panel B. Full decomposition of ΔR_{10Y} (%)				
	MP	FRN	CBI	RP
(1) Baseline (BBM ext. rest.)	26.4 [11.2, 43.2]	10.9 [0.1, 29.7]	3.2 [0.0, 15.0]	**59.5** [43.2, 73.5]
(2) BEX ext. rest. instead of BBM	26.4 [11.2, 43.2]	10.9 [0.1, 29.7]	3.2 [0.0, 15.0]	**59.5** [43.2, 73.5]
(3) No ext. rest.	26.4 [11.2, 43.2]	10.9 [0.1, 29.7]	3.2 [0.0, 15.0]	**59.5** [43.2, 73.5]
(4) − RP moves R_{10Y} more than MP	26.4 [11.6, 50.5]	10.9 [0.1, 29.7]	3.2 [0.0, 14.6]	**59.5** [37.6, 73.3]
(5) Scheduled meetings only	21.1 [12.2, 41.3]	7.1 [0.1, 24.3]	2.1 [0.0, 10.8]	**69.7** [47.8, 76.7]
(6) Statement windows only	21.4 [6.0, 42.0]	10.6 [0.2, 36.5]	2.5 [0.0, 15.1]	**65.5** [43.0, 73.5]
(7) Minutes releases added	23.5 [14.4, 43.9]	6.9 [0.0, 19.2]	4.2 [0.0, 17.8]	**65.3** [45.7, 74.3]

Notes. Rows (5) to (7) change the event set and so do not share the baseline's $\hat{\Sigma}$. Points and brackets follow Table 5; Panel B rows sum to 100 and bold marks the largest entry.

6 Impact of shocks on financial and macroeconomic indicators

6.1 Financial variables

To see how the shocks affect financial indicators, for each indicator y and each horizon h I estimate the local projection

$$\underbrace{y_{t+h} - y_{t-1}}_{\text{cumulative change}} = a_h + b_h' \varepsilon_t + u_{t,h}, \qquad h = 0, 1, \ldots, 30, \tag{16}$$

where t runs over FOMC announcement days. The dependent variable is the cumulative change in y from the end of the day before the announcement to the end of the day h business days after it, so $h = 0$ is the announcement-day response and $h = 30$ is the change over the following six weeks. The regressor ε_t collects the four structural shocks; on days with two announcement windows the shocks are summed. Each shock has unit variance, so b_h is the response to a one-standard-deviation shock in the units of y.

The regressors ε_t are estimated, and each accepted rotation satisfies all restrictions equally well. Different rotations produce different shock series and therefore different coefficients in the local projection. I summarize the uncertainty by a single distribution that combines rotation uncertainty across the identified set with estimation uncertainty in the local projections. To do that, I take the shocks from one accepted draw of the identification stage, estimate the local projection with heteroskedasticity- and autocorrelation-consistent standard errors, and repeat this for $K = 500$ accepted draws spaced evenly over the identified set, collecting $\{\hat{b}_h^{(k)},\ \widehat{\text{se}}(\hat{b}_h^{(k)})\}_{k=1}^{K}$. For each draw I then take $M = 30$ values from $N\big(\hat{b}_h^{(k)},\ \widehat{\text{se}}(\hat{b}_h^{(k)})^2\big)$, the normal approximation to the sampling distribution of the coefficient at that rotation, and pool the $K \times M$ values. The 90 and 68% bands are the 5th and 95th and the 16th and 84th percentiles of the pooled values, computed separately for each variable, shock, and horizon; the point estimate is the coefficient at the Fry–Pagan draw.

Figure 7 reports the responses of six financial indicators, one per row, to the four shocks, one per column: the two-year Treasury yield, the term premium and the expectations component of the ten-year Treasury yield from the decomposition of Adrian et al. (2013), the S&P500, the ICE BofA BBB corporate spread, and the risk appetite index of Bauer et al. (2023).

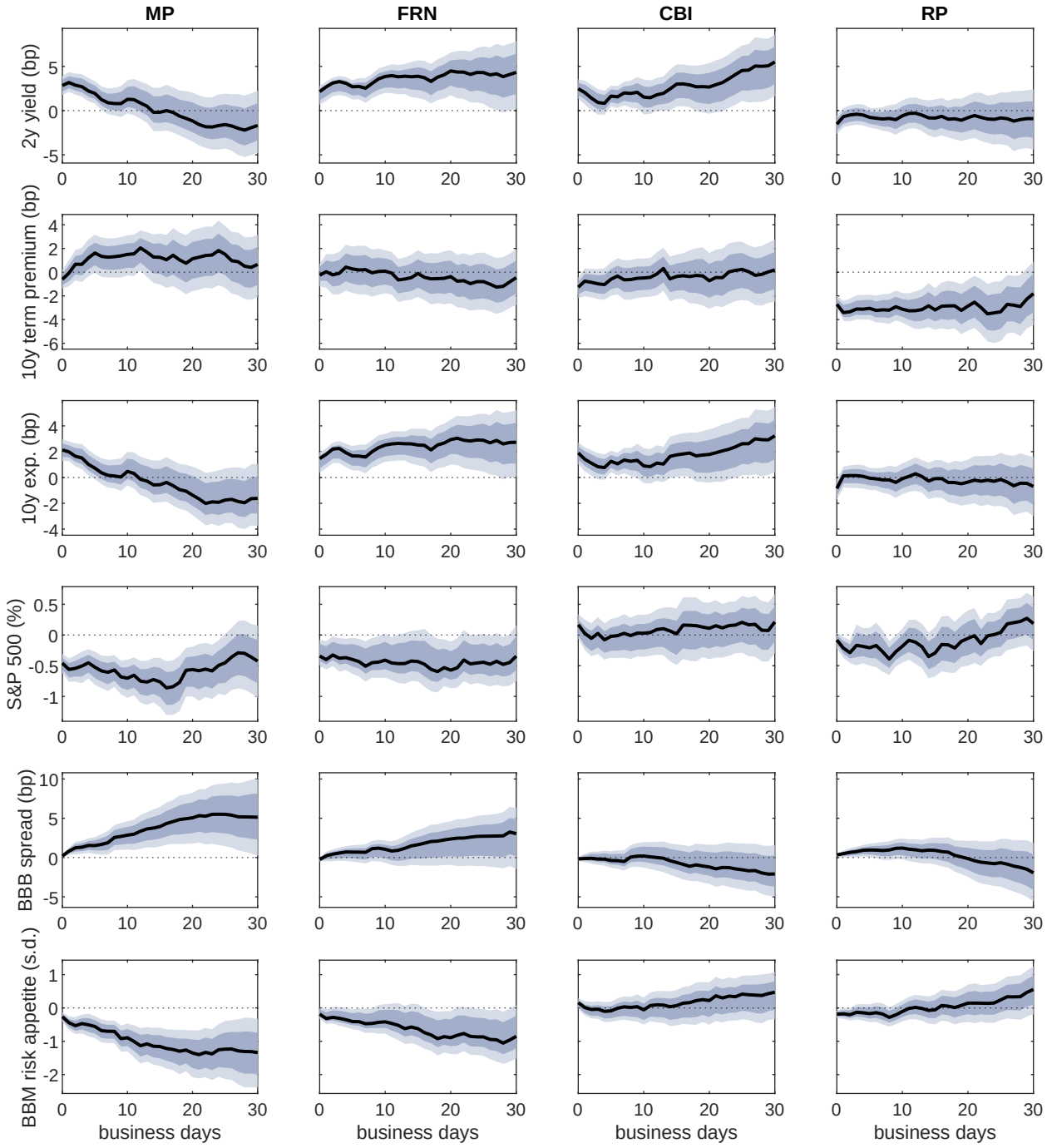


Figure 7: Daily responses to the four shocks

Notes. Solid lines are point estimates at the Fry–Pagan rotation; the light and dark shaded bands are the 90% and 68% ranges of the combined distribution described in the text. The discussion refers to the 90% band.

In each column of Figure 7 the responses of most financial indicators have the signs the shock's label implies. The monetary policy and Fed response to news shocks have standard contractionary effects on daily data: they raise the two-year yield on impact and lower equity

prices; over the following weeks both widen the BBB spread, by 3 to 5 basis points, and reduce measured risk appetite by about one standard deviation. The two-year yield response to the Fed response to news shock rises to about 4 basis points and persists through the end of the horizon, while the response to the policy shock fades within about three weeks. The central bank information shock raises the two-year yield as well, by about 5.5 basis points by the end of the horizon, and its effect on equity prices is small and not significant.

The risk premium shock lowers the ten-year term premium by about 3 basis points on impact, and the term premium stays lower throughout the six weeks. The two-year yield and the expectations component of the ten-year yield move by about 2 basis points or less, and not significantly after the announcement day. The other three shocks, in contrast, move the term premium by less, and rarely significantly, and operate more through the expectations component of the yield curve. The RP shock therefore moves the long end mainly through the compensation investors require for bearing duration risk rather than through the expected path of policy. This feature is not built into the identification: the restrictions require the risk premium shock to lower the ten-year yield, but they do not say which part of the yield moves. That is left to the data, and the data attribute most of the decline to the term premium.

The risk premium shock has small effects on equities, credit spreads and measured risk appetite, and they are not significant after the first week. In the first days after the announcement, the BBB spread widens by up to about 1 basis point, risk appetite falls by about 0.2 standard deviations, and stock prices fall by 0.2 to 0.4%. The label "risk premium" leads one to expect credit spreads and risk appetite to move; they move little, because the shock works mainly through the term premium. The restrictions identify it through the maturity structure of the yield response and leave its effect on risky assets free beyond the sign, and what they isolate is the part of the long-yield move that the expected policy path does not explain. A term premium is the compensation for bearing interest-rate risk, which Treasury and corporate bonds share: a change in it moves both yields together and leaves the spread unchanged.

Appendix 8.7 reports the same projections for the full set of variables, with a robustness specification that adds five lags of the daily change in the dependent variable. The two lines are close in most panels, and no response changes sign at a horizon where either band excludes zero.

6.2 Macroeconomic variables

To see how the shocks affect macroeconomic variables at monthly frequency, I estimate a monthly Bayesian VAR in the style of Jarociński and Karadi (2026). The VAR contains four shocks, ordered first, followed by five monthly variables: the one-year Treasury yield, the log S&P500, the excess bond premium of Gilchrist and Zakrajšek (2012), log real GDP, and the log PCE price index. The shocks enter as within-month sums of the high-frequency shocks, with zero in months without an announcement. The March 2020 announcements are excluded from the VAR, following Jarociński and Karadi (2026), and from the identification stage, so that the shocks are orthogonal over the announcements that enter the VAR. Within the VAR, identification is recursive with the shocks ordered first, so the shock columns do not depend on how the macro variables are ordered among themselves. The VAR has 12 lags and a Minnesota prior with overall tightness 0.2, estimated on 1991:01–2025:03. Each shock is scaled so that, at the Fry–Pagan rotation, it moves a Treasury yield by 25 basis points in the announcement window: the MP, FRN and CBI shocks raise the two-year yield, and the RP shock lowers the ten-year yield. As in the daily local projections, the constructed credible bands carry both parameter and identification uncertainty, over 2,000 draws that cycle through the 975 accepted rotations. The solid line corresponds to the center of the identified set in the sense of Fry and Pagan (2011).

Appendix Figure 18 reports the responses of all five variables to the MP, FRN and CBI shocks, and the RP shock is studied below in a larger VAR. The estimated impulse responses are largely consistent with macroeconomic theory and existing empirical evidence, with one exception, the one-year yield response to the monetary policy shock, which Appendix 8.8.1 discusses separately. Apart from that, a contractionary monetary policy shock produces the standard effects: stock prices fall, the excess bond premium rises, real GDP declines, and the price level falls persistently, with no price puzzle. A policy shock that raises the two-year yield by 25 basis points lowers stock prices by about 7% in the following month, raises the excess bond premium by about 0.3 percentage points on impact, and lowers real GDP by about 0.7% after ten months and the price level by about 0.7% after two years (Figure 8). The responses of stock prices and the price level are significant at the 68% level throughout, and that of real GDP from about half a year on. In the larger VAR used below for the RP shock, with nonresidential investment in place of total investment, nonresidential investment falls by about 1.9% after a year, significantly at the 68% level for most of the first two years.

A positive Fed response to news shock, a stronger response to public news than the market expected, raises the one-year yield persistently, in line with a systematic policy change. On the other variables, it has expansionary effects: stock prices rise, and activity and the price level increase, a temporary boom much like the one that follows an information shock. Since

growth is persistent, the good news that prompted the rate increase today keeps arriving for months, and the monthly responses reflect the news, not the policy move. A rate increase followed by a boom is what an econometrician would attribute to favorable information held by the Fed, which is why the response resembles the one to the information shock (Bauer and Swanson, 2023a; Jarociński and Karadi, 2026). A rate increase that is caused by the Fed's response to good public news therefore has the opposite effects to those of a policy shock: stock prices, activity and the price level rise. Jarociński and Karadi (2026), in contrast, find insignificant effects of FRN on economic activity, consistent with the stimulative effects of favorable fresh public news and the contractionary effects of tighter than expected systematic monetary policy broadly offsetting each other.

6.3 Comparison with the JK and CS systems

Figure 8 compares the responses to the monetary policy shock across the three systems: the JKCS response, with its bands, against the CS response in panel A and the JK response in panel B. The first row shows how the four shock series themselves respond. The shocks enter the VAR as monthly series, ordered first, so their responses show whether a shock in one month is followed by further shocks in later months. The second row shows the macroeconomic variables. In all three systems the policy shock raises the two-year yield by 25 basis points in the announcement window, so the lines are directly comparable.

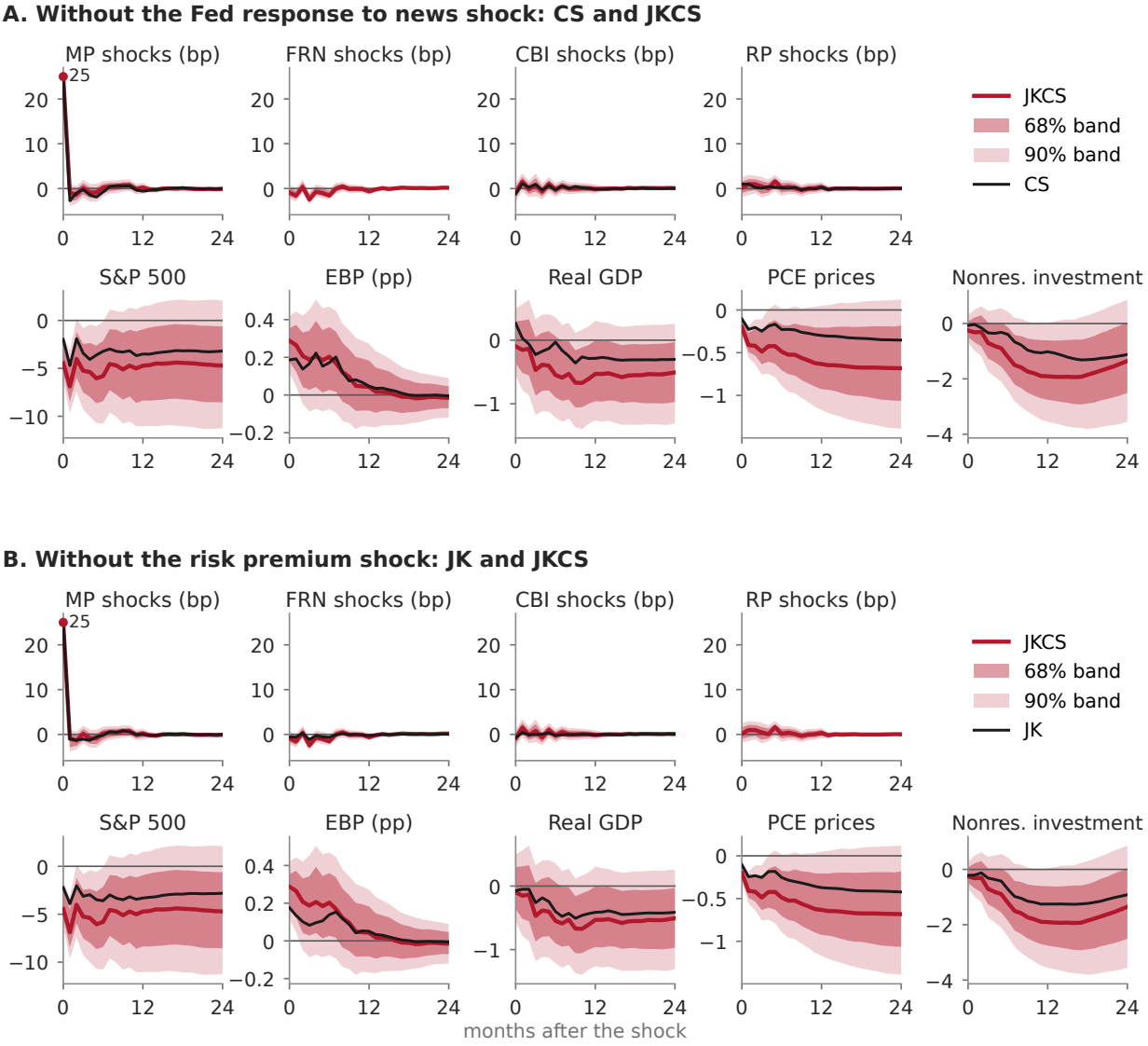


Figure 8: Monthly responses to the monetary policy shock, with and without the FRN and RP shocks

Notes. The policy shock raises the two-year yield by 25 basis points in the announcement window. Lines are responses at each system's Fry–Pagan rotation, the pointwise posterior median over VAR parameter draws; the 68 and 90% posterior bands are for JKCS. First row: monthly shock series, in basis points of the window rise in the two-year yield (MP, FRN, CBI) or fall in the ten-year yield (RP). Nonresidential investment is from the larger VAR of the next subsection, with nonresidential in place of total investment.

In the JKCS system the policy shock is not persistent: after the month of the shock, the policy series stays close to zero. The other three shock series also stay close to zero, except the Fed response to news series, which falls somewhat in the following months.

The CS and JKCS systems agree on the information shock but not on the policy shock,

and the difference matters for its effects. After the CS policy shock, real GDP and the price level fall by about 0.4% at the peak, against about 0.7% in the JKCS system, so the CS effects are 40–50% smaller. This is what the composition of the CS policy shock implies: about a quarter of it is the Fed response to news shock (Table 7), whose monthly effects are expansionary and offset part of the contraction. The same component explains why the CS policy series stays negative for several months after the shock. Omitting the Fed response to news shock therefore leads to underestimating the effects of monetary policy.

The JK effects are also smaller than the JKCS ones, by 25–45% at the peak, but the JK responses stay mostly inside the JKCS bands. The gap can come from the risk premium shock that the JK system omits or from the restrictions in which the two systems differ. The JK policy shock contains almost no risk premium variation, which points to the restrictions, and the restrictions also explain most of the gap between JK and JKCS in the window decomposition (Appendix 8.6). The omission of the risk premium shock matters instead for the information shock, about a tenth of which is risk premium variation in the JK system.

6.4 The risk premium shock

The risk premium shock does not contaminate the effects of monetary policy, but it has real effects of its own. To measure them, I add three monthly variables to the baseline VAR: the ten-year Treasury yield, log housing starts and log real private fixed investment. Lags, prior and sample are unchanged, and the RP shock is scaled to lower the ten-year yield by 25 basis points in the announcement window. Figure 9 reports the responses, including those of residential and nonresidential investment, estimated in the same VAR with either series in place of total investment.

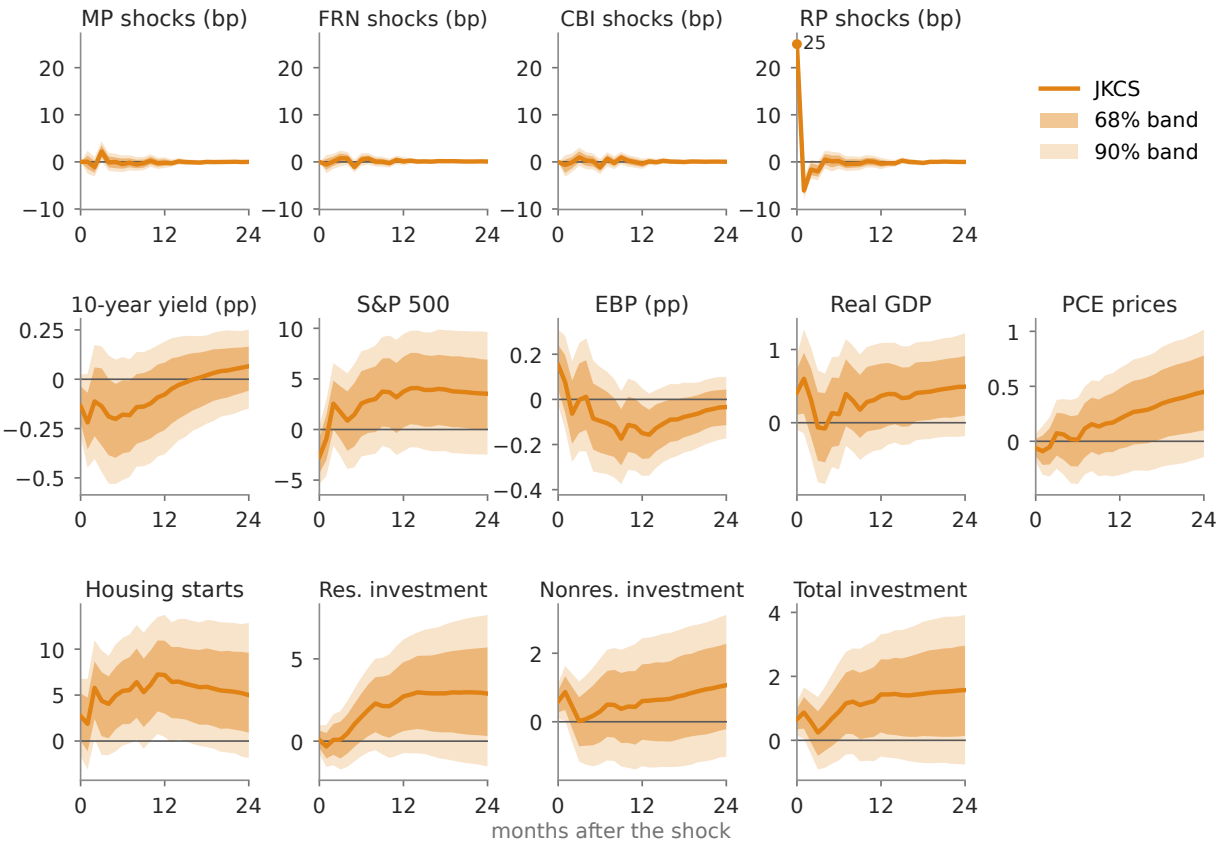


Figure 9: Monthly responses to the risk premium shock

Notes. The RP shock lowers the ten-year yield by 25 basis points in the announcement window. Lines are responses at the Fry–Pagan rotation, the pointwise posterior median over VAR parameter draws, with 68 and 90% posterior bands. First row: monthly shock series, in basis points as in Figure 8. Residential and nonresidential investment are from the same VAR with either series in place of total investment.

The risk premium shock is not persistent, but its effect on long-term rates is: the ten-year yield stays lower for about a year. In the month of the shock, stock prices fall and the excess bond premium rises, the pattern of a flight to safety; the identification does not restrict the excess bond premium. Real GDP and the price level do not respond significantly at the 90% level, although both are modestly higher in the second year.

The clearest real effect is on housing. Housing starts rise by about 7% after a year, and residential investment rises by about 3% and stays higher for two years. Nonresidential investment, in contrast, rises only briefly, in the month of the shock and the next. Total investment is about 1.5% higher after a year.

The pattern is consistent with the shock working through long-term rates. In daily data the risk premium shock lowers the ten-year yield mainly through the term premium, and in the monthly VAR the yield stays lower for about a year. That residential investment

rises persistently, and nonresidential investment only briefly, suggests a channel specific to housing, most likely mortgage finance: lower long-term yields reduce the cost of financing a home.

7 Conclusion

This paper proposes a high-frequency identification that recovers four shocks jointly at FOMC announcements: monetary policy, central bank information, Fed response to news, and risk premium shocks. The identification is based on sign and magnitude restrictions on how each shock moves bond and stock prices, derived from a small asset pricing model of the announcement window in the spirit of Cieslak and Schrimpf (2019). I estimate all four shocks on the same announcement windows and compare them with the three-shock systems of Jarociński and Karadi (2026) and Cieslak and Schrimpf (2019), to see where a shock that an identification omits ends up and whether it contaminates the policy shock.

First, omitting the Fed response to news shock, as in Cieslak and Schrimpf (2019), contaminates the policy shock. Without the Fed response to news shock, part of the Fed's predictable response to news enters the policy shock, and the effects of the policy shock on output and prices are underestimated. The jointly identified policy shock produces the conventional effects: stock prices fall, the excess bond premium rises, real GDP declines, and the price level falls.

Second, omitting the risk premium shock, as in Jarociński and Karadi (2026), barely contaminates the policy shock. However, part of what that system labels the information shock is, in fact, risk premium variation, although this part is small. The information effect survives the addition of the risk premium shock: the information shock still raises interest rates, stock prices and the price level and lowers the excess bond premium, as in Jarociński and Karadi (2020, 2026).

Third, the risk premium shock is important on its own. It is a long-end phenomenon that accounts for most of the variance of the ten-year yield surprise within announcement windows. It lowers the ten-year yield mainly through the term premium, and housing starts and residential investment rise after it. In the years of the asset purchase programs, the risk premium shock and the asset-purchase factor of Swanson (2021) capture largely the same variation.

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8 Appendix

8.1 Model appendix: bond loadings

This appendix derives the three yield loadings B_n^i, B_n^g, B_n^x used in Section 2.4 and establishes the two shape facts the identification relies on: $B_1^x = 0$, so the risk premium shock does not move the one-period yield, and $|B_n^x|$ peaks at a longer maturity than B_n^g does, so the premium loading is the one that dominates at the long end.

Let $P_t^{(n)}$ be the price of an n-period zero-coupon bond and $p_t^{(n)} = \log P_t^{(n)}$. Since $y_t^{(n)} = -\frac{1}{n} p_t^{(n)}$, write the log price as

$$p_t^{(n)} = -\bar{A}_n - \bar{B}_n' F_t, \qquad \text{so} \qquad y_t^{(n)} = \frac{\bar{A}_n}{n} + \frac{\bar{B}_n'}{n} F_t \equiv A_n + B_n' F_t.$$

Substituting the macro-factor equation into the policy rule and stacking the three equations of Section 2.1 gives the VAR(1) $F_{t+1} = \Phi F_t + \Sigma\, \varepsilon_{t+1}$, with

$$\Phi = \begin{pmatrix} \phi_g & 0 & 0 \\ a^{cb}\phi_g & \phi_i & 0 \\ 0 & 0 & \phi_x \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \sigma_g & 0 & 0 \\ a^{cb}\,\sigma_g & \sigma_i & 0 \\ 0 & 0 & \sigma_x \end{pmatrix}.$$

Prices are set under the market's model, in which a^m replaces a^{cb} in Φ and Σ. Holding the n-bond for one period gives $P_t^{(n)} = \mathbb{E}_t^m\big[M_{t+1}\, P_{t+1}^{(n-1)}\big]$, so $p_t^{(n)} = \log \mathbb{E}_t^m[e^{\xi}]$ with $\xi \equiv m_{t+1} + p_{t+1}^{(n-1)}$. The loadings follow from pricing the n-period bond one period at a time. Holding it for one period gives $P_t^{(n)} = \mathbb{E}_t^m\big[M_{t+1}\, P_{t+1}^{(n-1)}\big]$, so $p_t^{(n)} = \log \mathbb{E}_t^m[e^{\xi}]$ with $\xi \equiv m_{t+1} + p_{t+1}^{(n-1)}$. Guess that log prices are affine in the state, $p_{t+1}^{(n-1)} = -\bar{A}_{n-1} - \bar{B}_{n-1}' F_{t+1}$, and substitute this guess, the SDF and $F_{t+1} = \Phi F_t + \Sigma \varepsilon_{t+1}$. Then ξ splits into a part known at t and a shock:

$$\xi = \underbrace{-i_t - \tfrac{1}{2}\lambda_t'\lambda_t - \bar{A}_{n-1} - \bar{B}_{n-1}'\Phi F_t}_{\mathbb{E}_t^m[\xi]} \qquad \underbrace{- v'\varepsilon_{t+1}}_{\text{mean zero given } t}, \qquad v \equiv \lambda_t + \Sigma'\bar{B}_{n-1}.$$

Since ξ is normal given t, $\log \mathbb{E}_t^m[e^{\xi}] = \mathbb{E}_t^m[\xi] + \frac{1}{2}\operatorname{Var}_t(\xi)$, with

$$\tfrac{1}{2}\operatorname{Var}_t(\xi) = \tfrac{1}{2} v'v = \tfrac{1}{2}\lambda_t'\lambda_t + \lambda_t'\Sigma'\bar{B}_{n-1} + \tfrac{1}{2}\bar{B}_{n-1}'\Sigma\Sigma'\bar{B}_{n-1}.$$

The first term cancels the $-\frac{1}{2}\lambda_t'\lambda_t$ in $\mathbb{E}_t^m[\xi]$. The last term is a constant. The middle term varies with the state, because the price of risk does:

$$\lambda_t'\Sigma'\bar{B}_{n-1} = \lambda_{gx}\sigma_g D_{n-1}\, x_t, \qquad D_n \equiv \bar{B}_n^g + a^m \bar{B}_n^i,$$

where D_n is the bond's exposure to the priced growth shock. Collecting terms, the log price is

$$p_t^{(n)} = -\bar{A}_{n-1} + \tfrac{1}{2}\bar{B}_{n-1}'\Sigma\Sigma'\bar{B}_{n-1} - \big(e_i + \Phi'\bar{B}_{n-1} - \lambda_{gx}\sigma_g D_{n-1} e_x\big)' F_t,$$

using $i_t = e_i' F_t$ and $x_t = e_x' F_t$. This is affine in F_t, which confirms the guess, and matching the constant and the coefficient on F_t with $p_t^{(n)} = -\bar{A}_n - \bar{B}_n' F_t$ gives

$$\bar{A}_n = \bar{A}_{n-1} - \tfrac{1}{2}\bar{B}_{n-1}'\Sigma\Sigma'\bar{B}_{n-1}, \qquad \bar{B}_n = e_i + \Phi'\bar{B}_{n-1} - \lambda_{gx}\sigma_g D_{n-1}\, e_x.$$

Component by component, using $\Phi'\bar{B}_{n-1} = \big(\phi_g(\bar{B}^g_{n-1} + a^m\bar{B}^i_{n-1}),\ \phi_i\bar{B}^i_{n-1},\ \phi_x\bar{B}^x_{n-1}\big)'$,

$$\bar{B}^g_n = \phi_g\big(\bar{B}^g_{n-1} + a^m\bar{B}^i_{n-1}\big), \qquad \bar{B}^i_n = 1 + \phi_i\bar{B}^i_{n-1}, \qquad \bar{B}^x_n = \phi_x\bar{B}^x_{n-1} - \lambda_{gx}\sigma_g D_{n-1}.$$

Each loading has a clear source: $\bar{B}^i$ comes from discounting at the short rate, $\bar{B}^g$ from the expected response of future rates to growth, and $\bar{B}^x$ from the risk premium, which is nonzero only because λ_t moves with x_t. The starting values come from the one-period bond, which pays 1 and so has $P_t^{(1)} = \mathbb{E}_t^m[M_{t+1}] = e^{-i_t}$: hence $\bar{A}_1 = 0$ and $\bar{B}_1 = e_i$, that is $\bar{B}^i_1 = 1$ and $\bar{B}^g_1 = \bar{B}^x_1 = 0$. The one-period yield *is* the short rate, and it carries no premium. The same equation fixes the intercept μ_t of the log SDF in Section 2.2: $m_{t+1} = \mu_t - \lambda_t'\varepsilon_{t+1}$ is normal with mean μ_t and variance $\lambda_t'\lambda_t$, since ε_{t+1} has mean zero and identity covariance. For any normal ξ, $\mathbb{E}[e^\xi] = e^{\mathbb{E}[\xi] + \frac{1}{2}\operatorname{Var}(\xi)}$, so $\mathbb{E}_t^m[M_{t+1}] = e^{\mu_t + \frac{1}{2}\lambda_t'\lambda_t}$; equating the two expressions and taking logs gives $\mu_t = -i_t - \frac{1}{2}\lambda_t'\lambda_t$. With $S_n(\phi) \equiv 1 + \phi + \cdots + \phi^{n-1} = \frac{1-\phi^n}{1-\phi}$ the three recursions solve to

$$B^i_n = \frac{S_n(\phi_i)}{n} > 0, \qquad B^g_n = \frac{a^m\phi_g}{n}\,\frac{S_n(\phi_g) - S_n(\phi_i)}{\phi_g - \phi_i} \geq 0, \qquad B^x_n = -\frac{\lambda_{gx}\sigma_g}{n}\sum_{k=1}^{n-1}\phi_x^{n-1-k} D_k \leq 0,$$

with $D_k \equiv \bar{B}^g_k + a^m S_k(\phi_i) > 0$ and equality in the last two expressions only at $n = 1$, where $S_1(\phi_g) = S_1(\phi_i) = 1$ and the sum is empty.

Shapes. B^i_n is decreasing in n. B^g_n rises from $B^g_1 = 0$ to an interior peak and then declines toward zero. So does $|B^x_n|$, and two induction steps show why it peaks later. First, $\bar{B}^g_n$ is increasing, since $\bar{B}^g_n - \bar{B}^g_{n-1} = \phi_g(\bar{B}^g_{n-1} - \bar{B}^g_{n-2}) + a^m\phi_g(\bar{B}^i_{n-1} - \bar{B}^i_{n-2}) > 0$ from the base step $\bar{B}^g_2 - \bar{B}^g_1 = a^m\phi_g > 0$, so D_k is increasing. Second, writing $q_n \equiv -\bar{B}^x_n \geq 0$, the recursion $q_n = \phi_x q_{n-1} + \lambda_{gx}\sigma_g D_{n-1}$ with $q_1 = 0$ gives $q_n - q_{n-1} > 0$ by the same argument, so $|\bar{B}^x_n|$ is strictly increasing in n. Since D_k converges to $D_\infty = a^m/[(1-\phi_g)(1-\phi_i)]$, the log-price loading $\bar{B}^x_n$ converges to $-\lambda_{gx}\sigma_g D_\infty/(1-\phi_x)$ and the yield loading $B^x_n = \bar{B}^x_n/n$ therefore returns to zero, attaining an interior peak far out in n when x is near a unit root.

Two consequences for the identification. The peak of $|B_n^x|$ lies at or beyond the peak of B_n^g in n, and $|B_n^x|$ reaches its peak farther out in n with higher ϕ_x, other things equal. For example, at the calibration of Subsection 2.5, $\phi_g = 0.965$, $\phi_i = 0.9$ and $\phi_x = 0.9$, the peak of $|B_n^x|$ occurs at $n = 47$, roughly six years given eight announcements per year, against $n = 30$ for B_n^g. This delivers the restriction $|B_{N_{10}}^x| > |B_{N_2}^x|$: the risk premium shock affects longer-term rates more than shorter-term rates.

The two entries marked † in Table 1 follow. Writing N_2 and N_{10} for the two- and ten-year maturities, the CBI entry is $a^{cb}\big(B_{N_2}^i - B_{N_{10}}^i\big) > B_{N_{10}}^g - B_{N_2}^g$, whose left side is always positive and whose right side is negative whenever the hump in B_n^g peaks inside two years, in which case the inequality holds automatically. It can fail only when ϕ_g and ϕ_i are both near one. The RP entry is $|B_{N_{10}}^x| > |B_{N_2}^x|$, which holds when x is persistent and fails when x mean-reverts quickly.

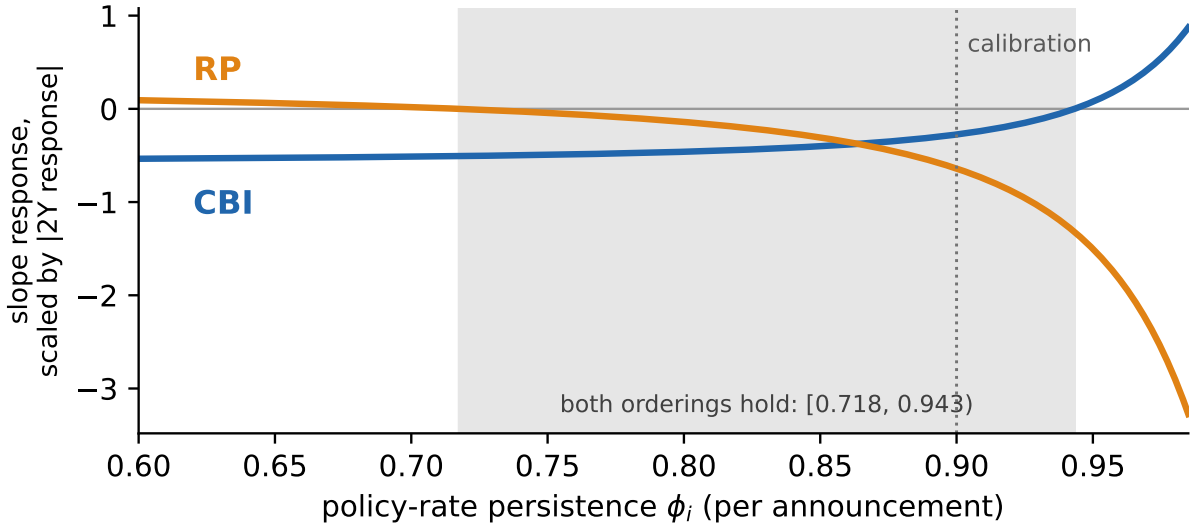


Figure 10: Admissible policy-rate persistence

Notes. The slope response (the ten-year minus the two-year response, scaled by the magnitude of the two-year response) is plotted against the policy-rate persistence ϕ_i, with the growth and price-of-risk persistences and γ^m/γ^{cb} held at their calibrated values. The shaded area marks the region where both † conditions hold, and the dotted line the calibration, $\phi_i = 0.9$. RP sets the lower bound and CBI the upper, leaving $\phi_i \in [0.718, 0.943)$.

8.1.1 Expected rates and the term premium

This subsection shows that the split of the yield response into an expected-rates part and a term premium is the split of B_n into the (B_n^i, B_n^g) part and the B_n^x part, so that the risk premium shock moves the premium and nothing else.

Iterate the policy rule forward under the market's model. The expected macro factor is $\mathbb{E}_t^m[g_{t+j}] = \phi_g^j g_t$, and the rule gives $\mathbb{E}_t^m[i_{t+j}] = \phi_i \mathbb{E}_t^m[i_{t+j-1}] + a^m \phi_g^j g_t$ with $\mathbb{E}_t^m[i_t] = i_t$. Neither the rule nor the growth process contains x, so the risk factor never enters: expected future short rates depend on i_t and g_t alone. Unrolling from $j = 0$,

$$\mathbb{E}_t^m[i_{t+j}] = \phi_i^j \, i_t + a^m \phi_g \, \frac{\phi_g^j - \phi_i^j}{\phi_g - \phi_i} \, g_t,$$

and averaging over $j = 0, \ldots, n-1$ with $\sum_{j=0}^{n-1} \phi^j = S_n(\phi)$ returns exactly the loadings of Appendix 8.1,

$$\frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}_t^m[i_{t+j}] = B_n^i \, i_t + B_n^g \, g_t, \qquad \text{so} \qquad TP_t^{(n)} = y_t^{(n)} - \left(B_n^i i_t + B_n^g g_t\right) = A_n + B_n^x \, x_t,$$

which depends on x alone. Since B_n^x is proportional to λ_{gx} (Appendix 8.1), setting $\lambda_{gx} = 0$ makes the term premium constant, and the expectations hypothesis holds. Differencing over the announcement window, with $\tilde{g}_t = \sigma_g \varepsilon_t^g$, $\tilde{x}_t = \sigma_x \varepsilon_t^x$ and A_n constant, splits the responses reported in Section 2.4 into the two parts,

$$\frac{1}{n} \sum_{j=0}^{n-1} \left(\mathbb{E}_t^m - \mathbb{E}_{t-\Delta}^m\right)[i_{t+j}] = B_n^i \, \tilde{\imath}_t + B_n^g \, \sigma_g \varepsilon_t^g, \qquad \widetilde{TP}_t^{(n)} = B_n^x \, \sigma_x \varepsilon_t^x,$$

so MP, CBI and FRN move only g and i, and therefore only expected rates.

8.2 Model appendix: stock loadings

This appendix solves the stock block and establishes three things: the price loadings B_s used in Section 2.4; that the risk premium shock is the only shock that moves the equity premium; and that it moves the equity premium and the term premium in *opposite* directions, which is what produces the stocks-down, yields-down signature the identification restricts.

Setup. Log dividend growth has conditional mean g_t and an independent cash-flow shock, $\Delta d_{t+1} = g_t + \sigma_d \varepsilon_{t+1}^d$ with $\varepsilon^d \perp (\varepsilon^g, \varepsilon^i, \varepsilon^x)$. The Campbell–Shiller identity, log-linearized around the mean log price–dividend ratio $\overline{pd}$, is $r_{s,t+1} = \kappa_0 + \kappa_1 pd_{t+1} - pd_t + \Delta d_{t+1}$ with $\kappa_1 = e^{\overline{pd}}/(1 + e^{\overline{pd}}) \in (0, 1)$. This is the one approximation in the model; the bond block is exact. Guess $pd_t = A_s + B_s' F_t$.

Solution. Pricing is under the market's model, so Φ and Σ are evaluated at a^m. The stock Euler equation $1 = \mathbb{E}_t^m[e^{m_{t+1}+r_{s,t+1}}]$ with a conditionally Gaussian exponent is $0 = \mathbb{E}_t^m[m_{t+1}+r_{s,t+1}]+\frac{1}{2}\operatorname{Var}_t[m_{t+1}+r_{s,t+1}]$. Collecting the terms in F_t and setting their coefficient to zero gives

$$B_s = \kappa_1 \Phi' B_s + e_g - e_i - \kappa_1 \lambda_{gx} \sigma_g D_s \, e_x, \qquad D_s \equiv B_s^g + a^m B_s^i,$$

where $-e_i$ comes from the SDF, e_g from dividends, and the last term from the cross moment $-\kappa_1 \lambda_t' \Sigma' B_s = -\kappa_1 \lambda_{gx} \sigma_g D_s \, x_t$ in the variance. Component by component, using $\Phi' B_s = (\phi_g D_s,\ \phi_i B_s^i,\ \phi_x B_s^x)'$, the three equations solve to

$$B_s^i = \frac{-1}{1-\kappa_1\phi_i} < 0, \qquad D_s = \frac{1 + a^m B_s^i}{1-\kappa_1\phi_g}, \qquad B_s^g = \kappa_1 \phi_g D_s + 1, \qquad B_s^x = \frac{-\kappa_1 \lambda_{gx} \sigma_g D_s}{1-\kappa_1\phi_x}$$

with $B_s^i < -1$, since $0 < \kappa_1 \phi_i < 1$.

The equity premium. Substituting $-i_t = \mathbb{E}_t^m[m_{t+1}] + \frac{1}{2}\operatorname{Var}_t(m_{t+1})$ into the Euler equation cancels every term that involves m alone, and the remaining terms give the premium:

$$EP_t \ \equiv \ \mathbb{E}_t^m[r_{s,t+1}] - i_t + \tfrac{1}{2}\operatorname{Var}_t\big(r_{s,t+1}\big) \ = \ -\operatorname{Cov}_t\big(m_{t+1}, r_{s,t+1}\big),$$

the expected excess log return plus the Jensen term. The return innovation is $(\kappa_1 \Sigma' B_s)' \varepsilon_{t+1} + \sigma_d \varepsilon_{t+1}^d$, so its loading on the priced shock is $\beta_g = \kappa_1 \sigma_g D_s$; since the SDF innovation is $-\lambda_{gx} x_t \varepsilon_{t+1}^g$ and the shocks are orthogonal with unit variance, every other term drops and

$$EP_t = \lambda_{gx} x_t \, \beta_g = \kappa_1 \lambda_{gx} \sigma_g D_s \, x_t, \qquad \widetilde{EP}_t = \kappa_1 \lambda_{gx} \sigma_g D_s \, \sigma_x \varepsilon_t^x, \qquad \widetilde{EP}_t\big|_{\text{MP, CBI, FRN}} = 0.$$

Three things follow. First, EP_t is proportional to x_t and to nothing else, so RP is the only shock that moves it. Second, the two premia move in opposite directions on that shock: $\widetilde{TP}_t^{(n)} < 0$ while $\widetilde{EP}_t > 0$. A positive RP shock raises the compensation demanded on equity, which is exposed to growth risk, and lowers it on bonds, which hedge growth risk. Stock prices fall while bond prices rise, the flight-to-safety co-movement the identification restricts. Third, the premium and the price loading on x are directly related: substituting for B_s^x gives $EP_t = -(1-\kappa_1\phi_x) B_s^x \, x_t$, so a unit rise in x raises the required return by $(1-\kappa_1\phi_x)|B_s^x|$ per period and lowers the log price by $|B_s^x|$, the capitalized value of those higher required returns.

Window response and signs. Within the window dividends are predetermined ($\varepsilon_t^d = 0$), so the surprise in the log price is the revaluation of pd, $\widetilde{\log P}_t = B_s' \widetilde{F}_t$ with $\widetilde{F}_t = (\sigma_g \varepsilon_t^g,\ \tilde{\imath}_t,\ \sigma_x \varepsilon_t^x)'$, which delivers the four responses of Section 2.4. Since $B_s^i < 0$, MP and FRN lower the price unconditionally. The other two signs are conditional, on different conditions. RP lowers the price if and only if $B_s^x < 0$, that is, if and only if

$$D_s > 0 \qquad \Longleftrightarrow \qquad a^m < 1 - \kappa_1 \phi_i,$$

which is also the condition under which the equity premium rises with the price of risk, since $EP_t = \kappa_1 \lambda_{gx} \sigma_g D_s \, x_t$. With the states in deviations from their means, the unconditional equity premium in the model is zero; a positive average premium would require a positive mean of x_t. That mean would enter only the constants A_n and A_s, not the loadings, so it would leave every window response unchanged, and the identification does not need it. CBI raises the price if and only if the cash-flow channel dominates the discount channel, $B_s^g + a^{cb} B_s^i > 0$, which after substitution and multiplication by $(1 - \kappa_1 \phi_g)(1 - \kappa_1 \phi_i) > 0$ is

$$(1 - \kappa_1 \phi_g)\, a^{cb} + \kappa_1 \phi_g \, a^m \ < \ 1 - \kappa_1 \phi_i,$$

the same bound imposed on a weighted average of the true and the perceived growth response. Since $a^{cb} > a^m$, the left side exceeds a^m, so the CBI condition is strictly tighter than the RP condition and implies it. This is why the CBI entry in Table 1 is the binding one. With $\kappa_1 \approx 1$ at announcement frequency, dividing both conditions by $1 - \phi_i$ gives $\gamma^m < 1$ for RP and $(1 - \phi_g)\, \gamma^{cb} + \phi_g \, \gamma^m < 1$ for CBI, where $\gamma^{cb} \equiv a^{cb}/(1 - \phi_i)$ and $\gamma^m \equiv a^m/(1 - \phi_i)$ are the true and perceived long-run responses of the policy rate to the macro factor.

8.3 Implementation of the restrictions

Let u_t be the $n \times 1$ vector of observables (high-frequency surprises) of a given system, demeaned over the T announcement windows, and let $\hat{\Sigma} = (T-1)^{-1} \sum_t u_t u_t' = PP'$ be its sample covariance matrix, with P the lower triangular Cholesky factor. Any orthogonal Q gives an observationally equivalent decomposition,

$$u_t \ = \ B \, \varepsilon_t, \qquad B = PQ', \qquad \varepsilon_t = B^{-1} u_t, \tag{17}$$

since $BB' = PQ'QP' = \hat{\Sigma}$ whenever $QQ' = Q'Q = I$. Identification is the choice of Q. The restrictions of Section 3 do not pin down a single Q: a set of rotations satisfies them, and the rejection algorithm below samples that set.

1. Draw an $n \times n$ matrix M with independent standard normal entries and take its QR decomposition, $M = QR$. If the ith diagonal entry of R is negative, replace the ith column of Q by its negative; Q is then uniform over the orthogonal matrices (Rubio-Ramírez et al., 2010). Form $B = PQ'$.

2. Normalize the sign of each column, for JKCS by the four diagonal entries of (11): $c_{11} > 0$, $c_{22} > 0$, $c_{33} > 0$, $c_{44} < 0$. A column and its negative describe the same decomposition, so this names the shocks and does not restrict the set.

3. Check the sign and magnitude restrictions, (11)–(14) for JKCS, (4) and (5) for JK, and (6)–(10) for CS, and discard at the first failure.

4. For JKCS, form $\varepsilon_t = B^{-1}u_t$ and keep the draw only if the risk premium shock satisfies (15). JK and CS impose no external restriction.

5. Repeat steps (1)–(4), storing every accepted B, for 10 million candidate rotations for JKCS, 5 million for CS and 3 million for JK, or until 100,000 draws are accepted.

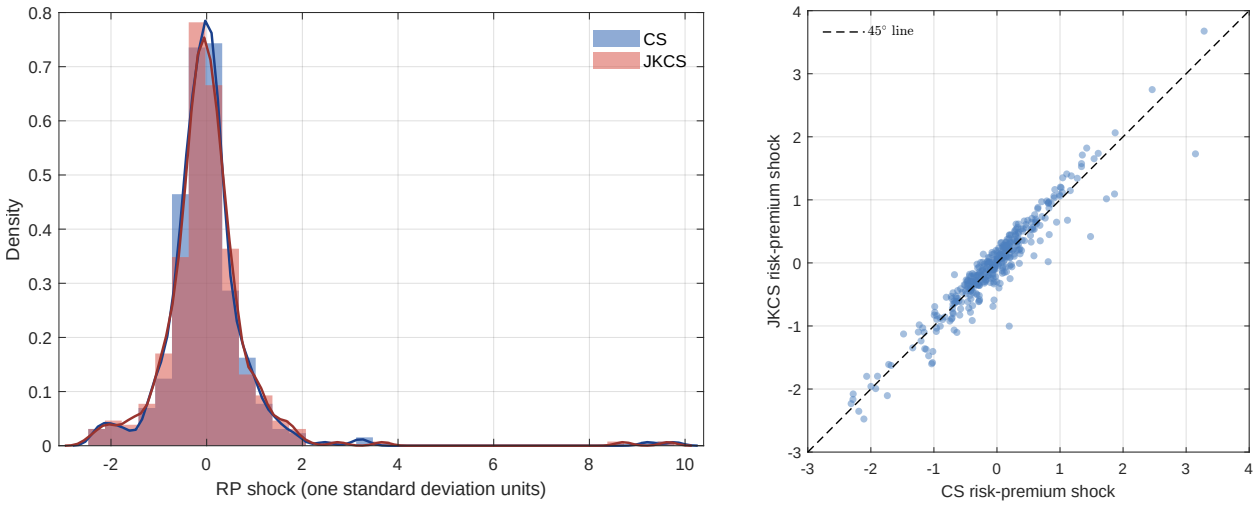
Point estimates for empirical exercises, unless stated otherwise, are taken at one accepted draw, the rotation of Fry and Pagan (2011),

$$k^* \;=\; \arg\min_{k \le K} \sum_{m=1}^{n} \sum_{j=1}^{n} \left(\frac{B_{mj}^{(k)} - \operatorname{med}_i B_{mj}^{(i)}}{\operatorname{sd}_i \; B_{mj}^{(i)}} \right)^2 , \tag{18}$$

where K is the number of accepted draws and the median and the standard deviation are taken over them entry by entry.

8.4 Risk premium shocks: comparison

Figure 11 compares the risk premium shock of the CS system with that of the four-shock system, both at the Fry–Pagan rotation of their own system: panel (a) plots the two distributions and panel (b) the two series against each other.



(a) Distributions (b) Scatterplot

Figure 11: The risk premium shock under the CS and the four-shock restrictions

Notes. Both series are at the Fry–Pagan rotation of their own system, on the same 382 announcement windows. Panel (a) plots the two distributions and panel (b) the two series against each other.

8.5 How much of the risk premium shock is delivered by the restrictions?

Most of the sign and magnitude restrictions on the fourth column of B come from the model's assumptions about the SDF and the policy rule, so the risk premium shock could in principle be an artifact of those assumptions rather than of the data. This appendix asks how much of it the restrictions deliver.

The exercise here estimates the same 4×4 system twice on the four observables of equation (11). The baseline imposes every restriction the model of Section 2 implies. The second estimation keeps all restrictions in the first three columns of B^{JKCS} and leaves the fourth column entirely free: no sign, magnitude, or external correlation restriction, except the sign of the ten-year loading, which is the normalization choice. Therefore, the baseline identified set is by construction a subset of the unrestricted one.

The shock in the fourth column does not disappear when the column is left free. In a square system three columns of B determine the fourth up to sign, since $B = PQ'$ and three rows of the orthogonal matrix Q fix the fourth, so the free column is whatever direction the other three leave, and there is no a priori reason for it to resemble the risk premium shock. Both estimations decompose the same $\hat{\Sigma}$ on the same $T = 382$ windows, so every difference between them comes from the identifying content placed on the fourth column, not from the observables, the sample, or the restrictions on the other three.

First, the restrictions do bind, and Table 13 shows how much and where. Taken together the eight conditions retain 8.1% of the RP-unrestricted set. Taken one at a time, the condition that the shock lowers equity prices removes about three fifths of the set of admissible rotations, and no other condition removes more than about two fifths: individual pass rates range from 38.3 to 94.5%.

Second, without the restrictions the center of the identified set is similar, except for the equity response, but the set is much less precise. Table 14 puts the two versions of the fourth column of the impact matrix B side by side. The unrestricted column still lowers the ten-year yield and the unpredictable component of the surprise, exactly as the baseline column does. The loading on the predictable component is close to zero in both. The magnitudes are attenuated but not qualitatively different. The median equity loading, in contrast, has the same size as the baseline one and the opposite sign.[7] Apart from that sign, what the restrictions change is the width of the identified set. The 5–95% bands are up to three times as wide without them. Finally, removing every restriction on RP leaves the other three shocks almost unchanged: the correlation between each of the unrestricted system's first three shocks and its baseline counterpart is 0.97 or above. The identifying content placed on the fourth column does not alter the first three shocks.

Table 13: Share of the RP-unrestricted set that satisfies each RP restriction

Restriction on RP	Condition	Share (%)
Lowers equity prices	$a_{3,4} < 0$	38.3
Lowers the unpredictable part of the surprise	$a_{1,4} < 0$	59.9
Lowers the two-year yield	$a_{1,4} + a_{2,4} < 0$	61.4
Moves against the risk-appetite index	$\mathrm{corr}(\varepsilon_t^{RP}, \Delta BBM_t) < 0$	67.1
Moves the ten-year more than MP does	$\lvert a_{4,1}\rvert < \lvert a_{4,4}\rvert$	70.3
Loads more on the unpredictable part	$\lvert a_{2,4}\rvert < \lvert a_{1,4}\rvert$	86.9
Moves the ten-year more than the two-year	$\lvert a_{1,4} + a_{2,4}\rvert < \lvert a_{4,4}\rvert$	91.9
Moves the two-year less than MP does	$\lvert a_{1,1} + a_{2,1}\rvert > \lvert a_{1,4} + a_{2,4}\rvert$	94.5
All eight jointly		8.1

Notes. Of 10 million rotations, 12,048 satisfy every restriction on MP, FRN and CBI. Each row reports the share of those 12,048 that also satisfy the stated RP restriction, taken one at a time; the last row reports the share satisfying all eight at once. Rows are ordered from most to least binding. The ten-year sign is the column normalization and is therefore not testable here.

[7] Absent the restrictions, 38.3% of admissible rotations have a risk premium shock that lowers equity prices.

Table 14: Impact loadings of the RP column, with and without its restrictions

	Baseline	RP unrestricted
Row of B	median [5, 95]	median [5, 95]
RES	−0.010 [−0.020, −0.002]	−0.004 [−0.027, 0.020]
FIT	−0.001 [−0.006, 0.004]	−0.000 [−0.006, 0.005]
Δ S&P500	−0.061 [−0.188, −0.004]	0.061 [−0.220, 0.334]
ΔR_{10Y}	−0.032 [−0.036, −0.028]	−0.030 [−0.040, −0.014]

Notes. Point estimates are the element-wise median across the identified set, not a single admissible rotation; brackets are the fifth and ninety-fifth percentiles across the set. Units are percentage points for RES, FIT and the ten-year yield, and % for the S&P500, so magnitudes are comparable within a row but not across rows. The ten-year sign is imposed as the column normalization in both columns of the table.

8.6 Isolating the cost of an omitted shock

The three-shock systems in Section 5 differ from the baseline in two ways at once: they identify one shock fewer, and they impose only a subset of the baseline restrictions. For example, the risk premium shock here must correlate negatively with the daily risk-appetite index of Bauer et al. (2023), which Cieslak–Schrimpf do not require, and the information shock must move RES more than FIT, which Jarociński–Karadi do not require. Thus, it is not clear whether the difference in results is due to the added shock or to the difference in restrictions.

This appendix separates the two by varying only the restrictions: each three-shock system is re-estimated under the restrictions of the four-shock system rather than its published ones. Because RES + FIT is the two-year surprise, both three-shock observable sets are nested in the baseline's. Cieslak–Schrimpf give up the split of the policy surprise into RES and FIT, which separates FRN from MP, and Jarociński–Karadi give up the ten-year yield, which identifies RP. Restrictions that refer to what a system gives up cannot be imposed on it, but the rest carry over unchanged. Each re-estimated system therefore differs from the baseline in the missing shock alone, which is what makes the exercise possible.

Formally, write s^S_{mj} for the variance share of variable m explained by shock j, defined in Section 5, with a superscript for the system: $S = P$ for a three-shock system as published, $S = P^*$ for the same system re-estimated under this paper's restrictions, and $S = JKCS$ for the baseline. For every observable m and every shock j the two systems share,

$$\underbrace{s^{\text{JKCS}}_{mj} - s^{P}_{mj}}_{\text{the gap of Section 5}} = \underbrace{s^{P^*}_{mj} - s^{P}_{mj}}_{\text{restrictions}} + \underbrace{s^{\text{JKCS}}_{mj} - s^{P^*}_{mj}}_{\text{missing shock}}. \tag{19}$$

Tables 15 and 16 report the results. Against Cieslak–Schrimpf, swapping the restrictions moves no share by more than two percentage points, while the shock omission moves them by up to twenty-five: the gaps in Section 5 are therefore the price of the missing shock. Against Jarociński–Karadi the opposite holds. The four-shock system adds several conditions on the three shocks Jarociński–Karadi identify, including the one mentioned above, and they move the shares by more than the missing shock alone does. Thus, those gaps cannot be read as the cost of the shock omission alone.

Table 15: Variance-share gaps against Cieslak–Schrimpf, split into restrictions and the missing shock

Variable	MP	CBI	RP
Restrictions			
ΔR_{2Y}	+0.7	-0.2	-0.4
Δ S&P500	-0.4	+0.3	+0.0
ΔR_{10Y}	+0.6	+0.2	-0.8
Missing shock			
ΔR_{2Y}	-13.2	-6.2	-0.7
Δ S&P500	-24.9	+3.9	-0.2
ΔR_{10Y}	-9.4	-0.8	-0.8

Notes. The three-shock system is estimated twice on the same observables and with the same shocks, once under the restrictions Cieslak–Schrimpf impose and once under those restrictions of the four-shock system that its observables and shocks allow. Entries are announcement-window variance shares in percentage points. Each system is evaluated at its own Fry–Pagan rotation, as in Section 5, and the two blocks add to the gap reported there.

Table 16: Variance-share gaps against Jarociński–Karadi, split into restrictions and the missing shock

Variable	MP	FRN	CBI
Restrictions			
RES	-11.8	+0.1	+11.7
FIT	+0.6	+7.3	-7.9
Δ S&P500	+12.8	-6.5	-6.3
Missing shock			
RES	-14.1	+2.7	+8.7
FIT	+6.3	-1.5	-5.3
Δ S&P500	+0.8	+4.4	-6.9

Notes. The three-shock system is estimated twice on the same observables and with the same shocks, once under the restrictions Jarociński–Karadi impose and once under those restrictions of the four-shock system that its observables and shocks allow. Entries are announcement-window variance shares in percentage points. Each system is evaluated at its own Fry–Pagan rotation, as in Section 5, and the two blocks add to the gap reported there.

8.7 Daily responses of financial variables

Figures 12–15 report the daily local projections described in Section 6.1 for thirteen variables, one figure per shock and one panel per variable, with units given in each panel title; the two-year yield is in Figure 7 only. The sample keeps every announcement with the required daily data and the horizon of projections is set to 30 business days; the number of announcements retained differs across panels because the underlying series begin and end on different dates. The solid line is the specification without controls, the dashed red line adds five lags of the daily change in the dependent variable, and the shaded band is the 5–95% range of the identified set combined with sampling uncertainty in the projection, for the specification without controls; the dotted lines trace the same range for the five-lag specification. The CDX series are total-return indices, so a *rising* line means credit is performing well, that is, risk on. BEX is the Bekaert–Engstrom–Xu index of risk aversion, so a rising line is a move toward risk off, the opposite sign convention to the Bauer et al. (2023) index of risk appetite. With the lags, no point estimate leaves the band of the specification without them, and no response changes sign at a horizon where either band excludes zero, so the results do not depend on the controls.

Where the responses are significant at the 90% level, the policy and Fed response to news shocks raise yields, the VIX, the BBB spread and BEX risk aversion and lower breakeven rates, CDX returns, stock prices and risk appetite, the directions of a monetary contraction.

The risk premium shock moves the VIX, the BBB spread, BEX risk aversion, CDX returns, stock prices and risk appetite in the same directions but lowers Treasury yields, the pattern of a flight to safety. Its only other significant response is that of the two-year breakeven rate, which rises by about 2 to 3 basis points on the announcement day and at times over the following three weeks. The information shock raises yields; apart from that, only its term premium response, negative on impact, and its CDX investment-grade return, positive at the end of the horizon, are significant.

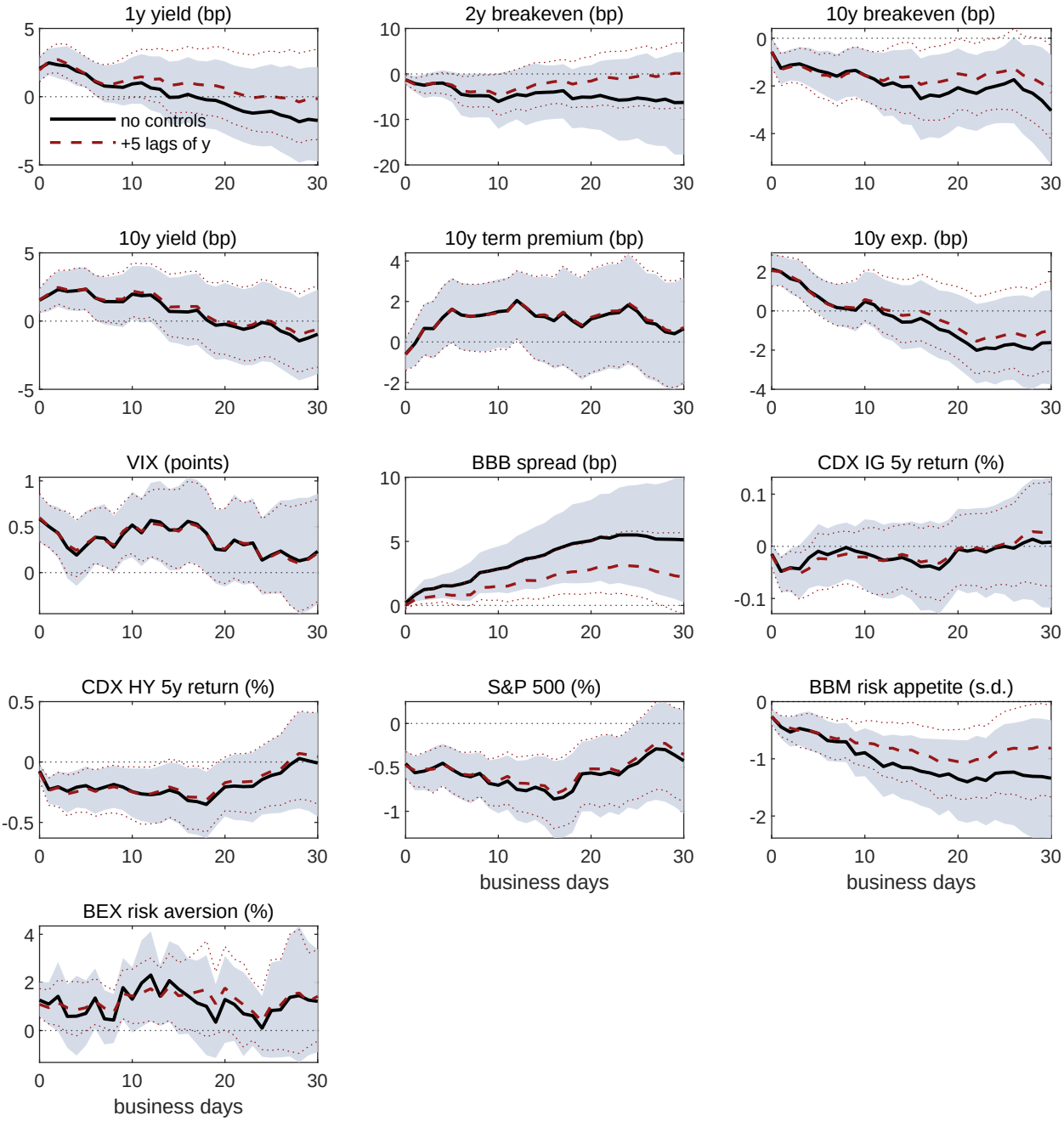


Figure 12: Daily responses to the monetary policy shock

Notes. Lines, bands and sample follow the description in the text; units are given in each panel title.

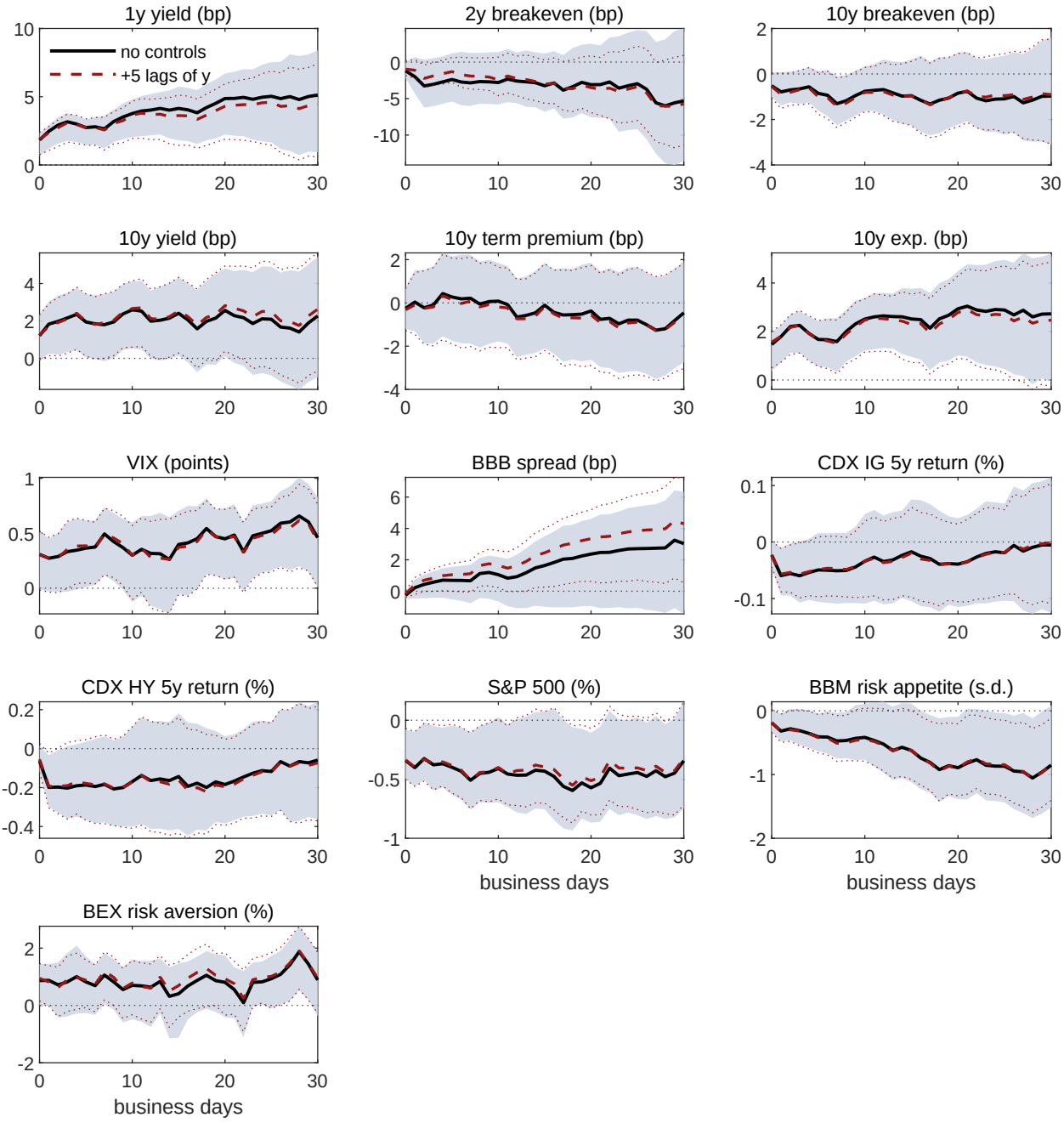


Figure 13: Daily responses to the Fed response to news shock

Notes. Lines, bands and sample follow the description in the text; units are given in each panel title.

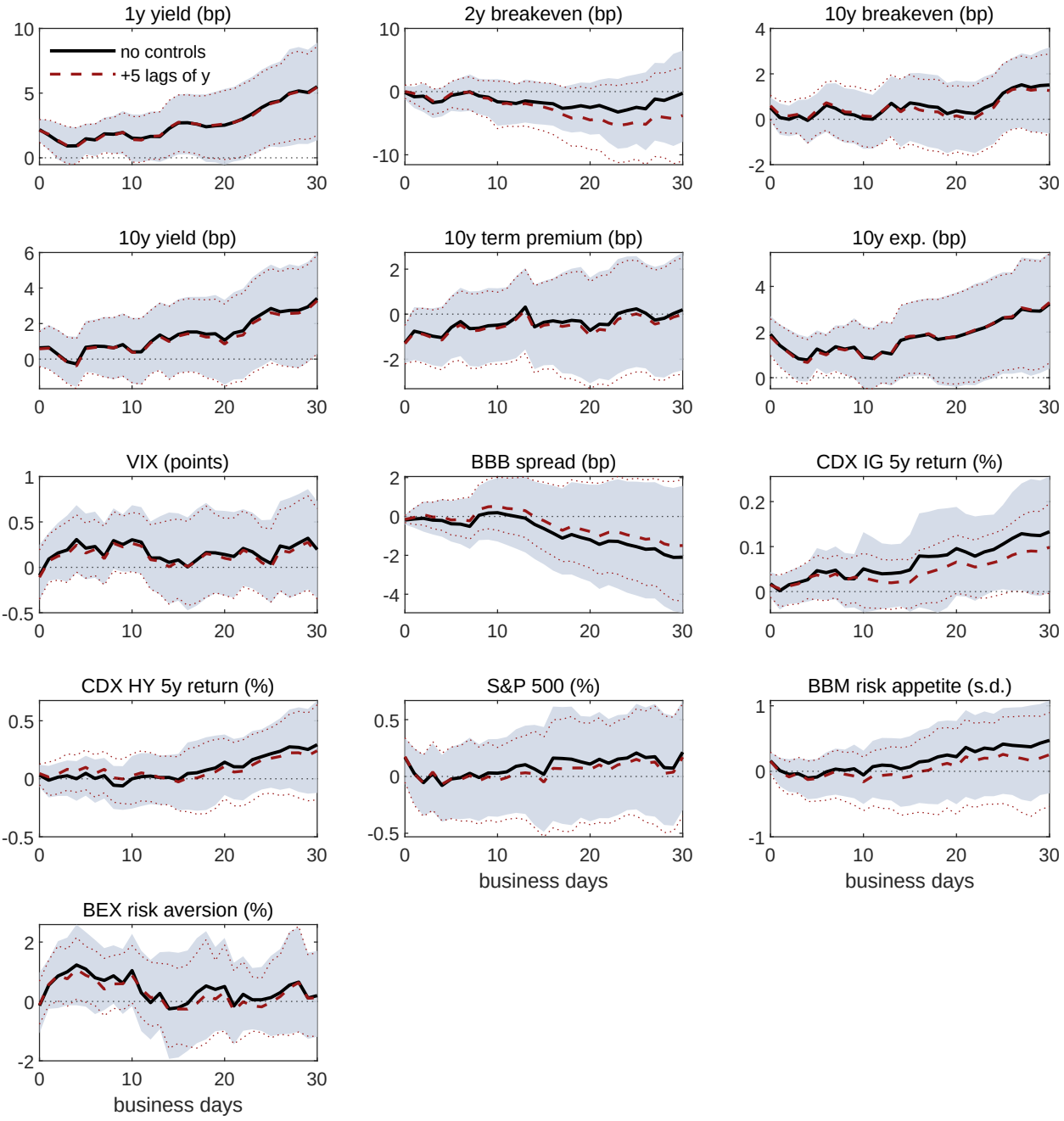


Figure 14: Daily responses to the central bank information shock

Notes. Lines, bands and sample follow the description in the text; units are given in each panel title.

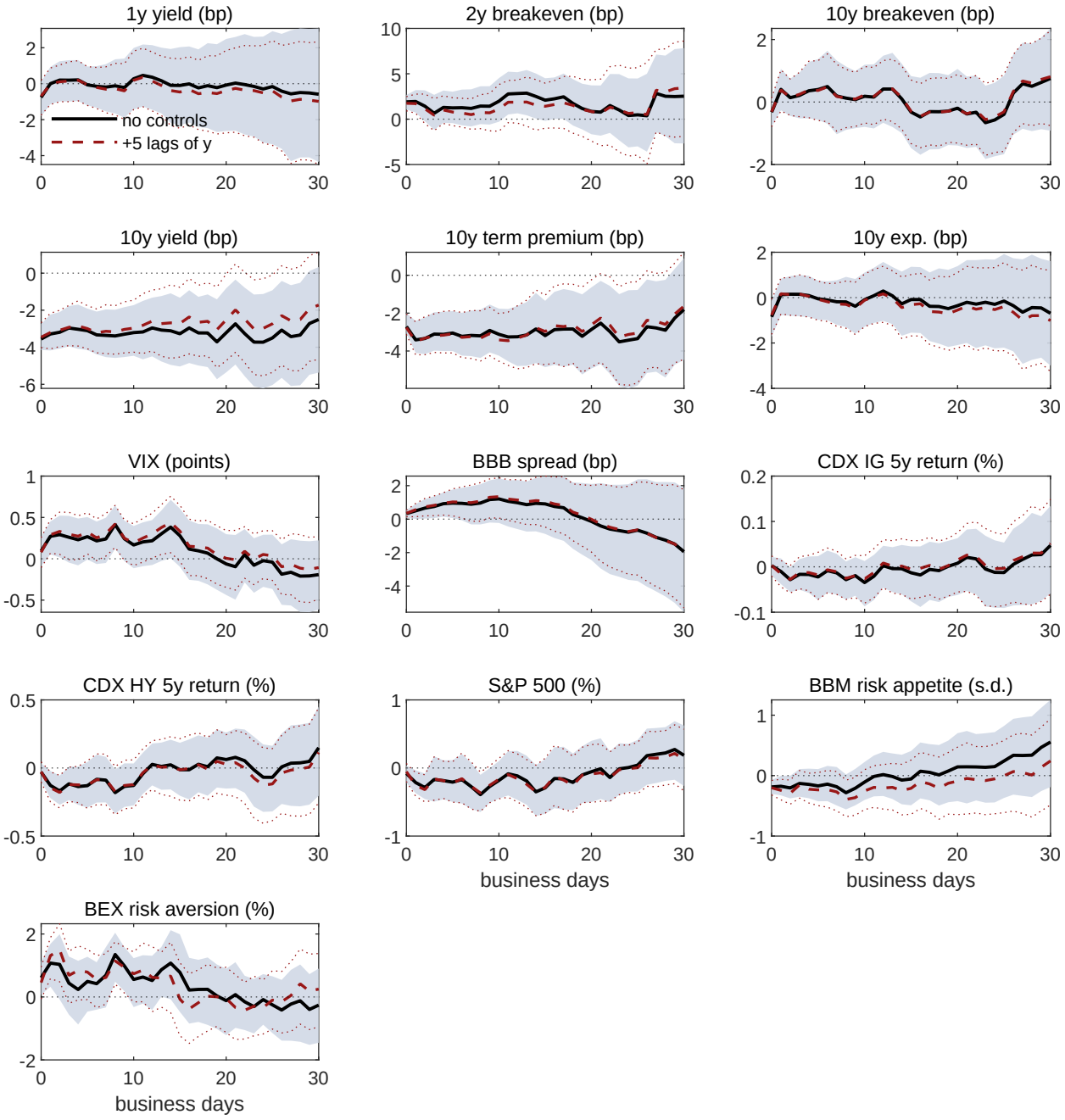


Figure 15: Daily responses to the risk premium shock

Notes. Lines, bands and sample follow the description in the text; units are given in each panel title.

What the projections look like with one shock missing

Figures 16 and 17 ask how the daily responses of financial variables change when one of the four shocks is not identified. For each figure, I estimate equation (16) twice, with the four-shock system's shocks and with a three-shock system's shocks, and plot the two responses together. Each shock of a three-shock system is an exact combination of the four JKCS

shocks, with the shares in Table 7. The three-shock systems also keep their own restrictions, so a gap between the lines can also come from the difference in restrictions.

Omitting the Fed response to news shock changes mainly the responses to the monetary policy shock. In Figure 16 the responses to the information and risk premium shocks stay inside the 68% bands of their four-shock counterparts throughout. The three-shock policy shock raises the two-year yield and the expectations component of the ten-year yield by more and for longer: the two-year yield is about 3.1 basis points higher after two weeks, against 1.3 in the four-shock system, and the three-shock line lies above the four-shock 68% band for most of the horizon. It also lowers stock prices, widens the BBB spread and lowers risk appetite by 13 to 22% more at the peak. The difference comes from the omitted shock: 26.0% of the three-shock policy shock is the Fed response to news shock (Table 7), which raises the two-year yield by about 4 basis points after two weeks and has the same effects on risky assets on its own.

Without the risk premium shock (Figure 17), the responses to the information shock change the most. The three-shock information shock lowers the ten-year term premium by about 0.2 basis points on impact, against 1.3 in the four-shock system, a difference significant at the 90% level in the first two days. It also lowers the BBB spread and raises stock prices and risk appetite by more than the four-shock information shock, and the three-shock line lies outside the 68% band for much of the first three weeks. Of the three-shock information shock, 79.7% is the four-shock information shock, 9.1% the risk premium shock and 8.4% the policy shock (Table 7); the last two enter with negative signs, which raise the term premium on impact and move risky assets toward risk on.

For the policy and Fed response to news shocks, the responses of stock prices, the BBB spread and risk appetite change by 6% or less at the peak, and those of the two-year yield and the expectations component by more: on impact, the three-shock policy shock raises the two-year yield by about 3.7 basis points, against 2.8 in the four-shock system, and the three-shock Fed response to news shock by about 1.5, against 2.1. Both figures compare point estimates. Apart from the term premium response to the information shock in the first two days, every three-shock response lies inside the 90% band of its four-shock counterpart throughout.

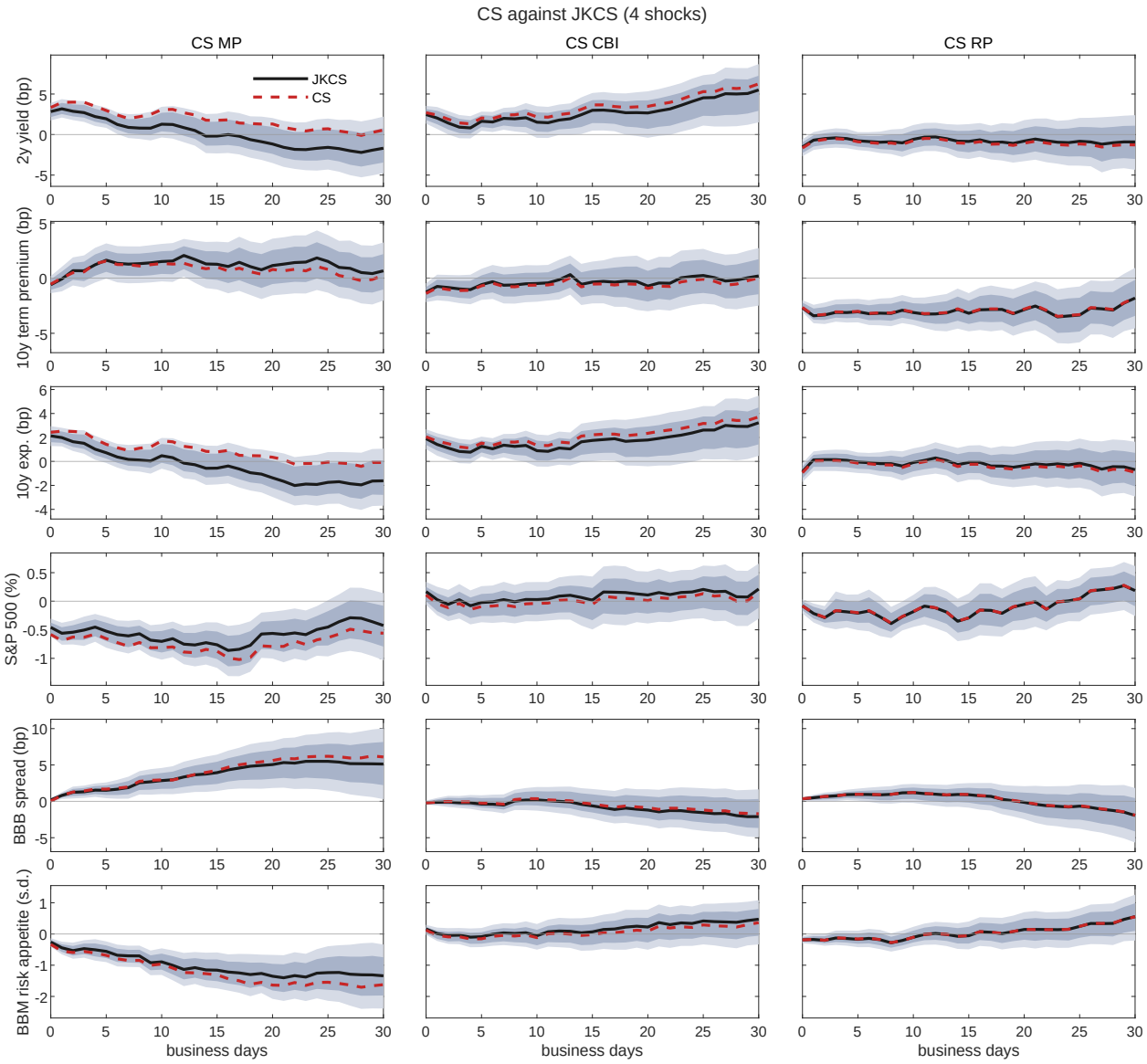


Figure 16: Daily responses with and without a Fed response to news shock

Notes. Rows are variables and columns are the three shocks of the three-shock system. The solid line and the shaded bands are the four-shock system of Figure 7; the dashed line is the three-shock system. Both are estimated on the same announcement days with the same specification.

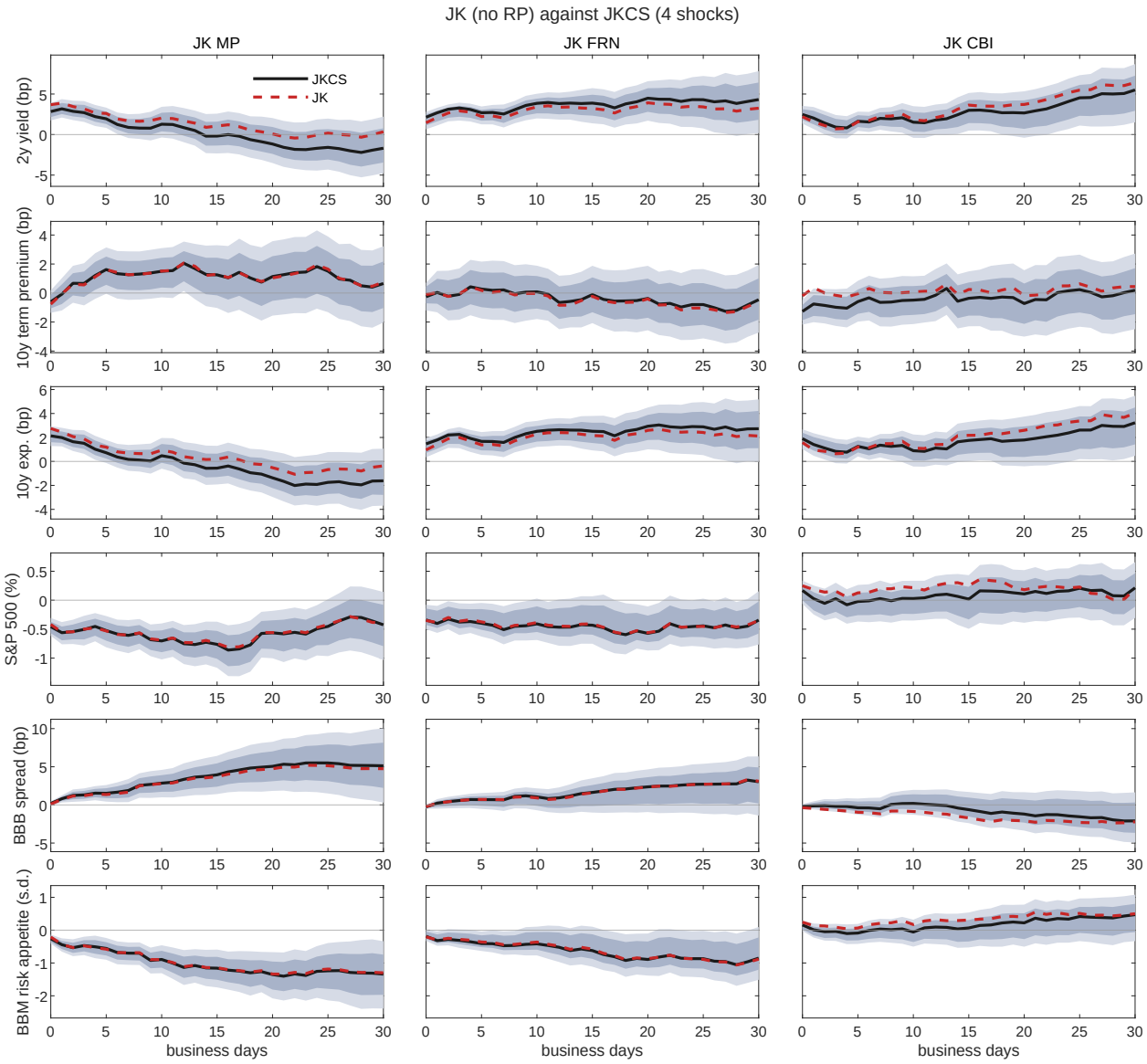


Figure 17: Daily responses with and without a risk premium shock

Notes. Layout as in Figure 16.

8.8 Monthly VAR: responses of all variables

Figure 18 reports the responses of the five variables of the VAR in Section 6.2 to the MP, FRN and CBI shocks, scaled as in the main text. Section 6.4 reports the RP shock.

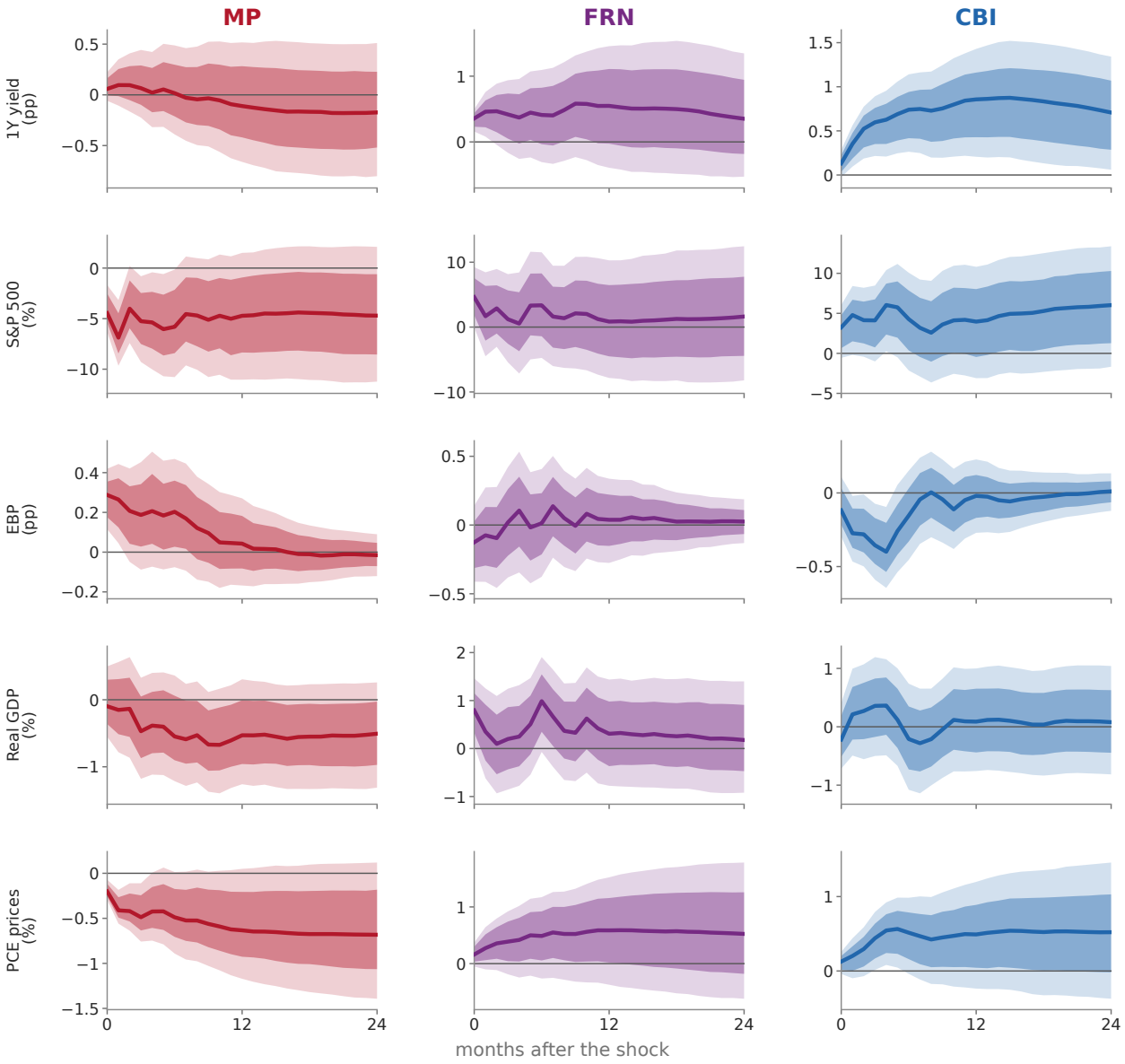


Figure 18: Monthly BVAR impulse responses to the MP, FRN and CBI shocks

Notes. Columns: monetary policy (MP), Fed response to news (FRN), and central bank information (CBI) shocks; rows: the one-year Treasury yield, log S&P500, excess bond premium (EBP), log real GDP, and log PCE price index. Each shock raises the two-year yield by 25 basis points in the announcement window at the Fry–Pagan rotation. Solid lines are posterior medians at the Fry–Pagan rotation; shaded areas are 68 and 90% posterior bands from 2,000 draws over the accepted rotations.

8.8.1 The one-year yield response to the monetary policy shock

The one-year yield response to the monetary policy shock in Figure 18 is negative at longer horizons and never significant at the 90% level, while every other row of the MP column

shows the standard contractionary effect of the shock. Such a puzzling response is not specific to this system: on the same announcements, the JK and CS policy shocks show the same response. It is also visible in the daily local projections: despite a positive effect on the yield in the first two weeks, the reaction eventually becomes insignificant.

The same feature is discussed in Jarociński and Karadi (2026), and their estimates identify its source. On FOMC announcements alone, with sign restrictions only, which is the design of this paper, their monetary policy shock does not move the one-year yield. In their baseline, which has a larger event set (speeches, testimonies and minutes releases) and additional heteroskedasticity-based restrictions, the reaction of the one-year rate is conventional. Thus, the reason for the puzzling reaction lies either in the event set or in the heteroskedasticity-based part of the identification. Its investigation is left for future work.